Learning Firm Conduct: Pass-Through as a Foundation for Instrument Relevance*

Adam Dearing† Lorenzo Magnolfi‡ Daniel Quint§

Christopher Sullivan¶ Sarah Waldfogel‖

April 9, 2026

Abstract

Understanding firm behavior is essential to detecting collusion, designing tax policy, quantifying markups, and many other objectives in economic research. Leading empirical approaches use instrumental variables (IV) to evaluate models of firm conduct. We provide a formal econometric framework that bridges between modern structural IV methods and classic reduced-form intuition, illuminating when and why instruments actually have the power to distinguish competing conduct models. Model-implied cost pass-through emerges as a key determinant of instrument relevance, and different instruments target distinct features of cost pass-through. Practitioners can use our framework to select instruments tailored to their specific economic question – e.g., a counterfactual policy evaluation. We apply these insights to Washington's marijuana market, where we test conduct with instruments targeted toward computing the Laffer curve. The ubiquitous Bertrand-Nash model is rejected in favor of a "Keystone" rule-of-thumb where firms price at twice marginal cost, with substantial implications for the Laffer curve.

Keywords: Testing conduct, pass-through, instrument relevance, falsification

*Some results first appeared in the working paper "Falsifying Firm Conduct" (Magnolfi, Quint, Sullivan, and Waldfogel, 2022a), which this paper supersedes. We thank Anya Tarascina for tremendous research assistance, and Brett Hollenbeck, Nathan Seegert, and Kosuke Uetake for kindly sharing data with us. We also thank Daniel Ackerberg, Matthew Backus, Steven Berry, Christopher Conlon, Mert Demirer, Jeremy Fox, Paul Grieco, Philip Haile, Justin Johnson, Nathan Miller, Conor Ryan, Katja Seim, and Matthew Weinberg for helpful discussions, as well as numerous seminar and conference participants. Any errors are our own.

†Samuel Curtis Johnson Graduate School of Management, Cornell University and NBER. Email: aed237@cornell.edu

‡Department of Economics, University of Wisconsin-Madison. Email: magnolfi@wisc.edu

§Department of Economics, University of Wisconsin-Madison. Email: dquint@ssc.wisc.edu

¶Department of Economics, University of Calgary. Email: christopher.sullivan1@ucalgary.ca

‖Darla Moore School of Business, University of South Carolina. Email: sarah.waldfogel@moore.sc.edu

1 Introduction

Understanding firm conduct – a key feature of structural equilibrium models – is essential for economists. For example, the demand-conduct approach to inferring firms' price-cost markups over time requires an assumption on firm conduct (e.g., Atalay, Frost, Sorensen, Sullivan, and Zhu, 2025; Grieco, Murry, and Yurukoglu, 2023; Döpper, MacKay, Miller, and Stiebale, 2025).[1] Firm conduct also plays a key role in counterfactual analysis; e.g., optimal tax policy computations depend on assumptions about firm behavior (Miravete, Seim, and Thurk, 2018; Hollenbeck and Uetake, 2021).

Beyond its role as a crucial model primitive, firm conduct in and of itself is an object of direct interest for economists. Detecting whether firms are colluding is a classic objective in the industrial organization (IO) literature, with recent work examining the pharmaceutical industry (Starc and Wollmann, 2025), labor markets (Roussille and Scuderi, 2026), and industries in developing economies (Sharma, 2025), along with growing interest due to increasing adoption of pricing algorithms leveraging AI and machine learning (e.g., Calder-Wang and Kim, 2024).

Researchers have developed various methods to infer firm conduct from data. An influential early approach uses pass-through regressions to distinguish conduct models based on how firms pass cost shocks through to prices (Sumner, 1981). While intuitive, this approach only works for a narrow class of models and requires specific cost shifters like unit taxes (Bulow and Pfleiderer, 1983; MacKay, Miller, Remer, and Sheu, 2014).

A more general approach, pioneered by Bresnahan (1982) in homogeneous product markets and formalized by Berry and Haile (2014) in modern differentiated product environments, uses instrumental variables (IV) to test conduct through exclusion restrictions. To have bite and result in valid inference on conduct, these restrictions must be based on instruments that are *relevant for testing* (Duarte, Magnolfi, Sølvsten, and Sullivan, 2024), which requires instruments to be correlated with differences in competing conduct models' implied markups (Berry and Haile, 2021; Backus, Conlon, and Sinkinson, 2021). However, the fundamental drivers of this correlation are unclear, since there are multiple channels through which instruments can correlate with markups. Without understanding these drivers, researchers risk selecting instruments that appear valid but provide no power to distinguish between competing conduct models—potentially leading to inconclusive tests or, worse, spurious con-

[1] The demand-conduct approach complements the alternative "production" approach (De Loecker and Warzynski, 2012; De Loecker, Eeckhout, and Unger, 2020), which typically requires an assumption of cost-minimizing behavior by firms. For a comparison between the two approaches, see De Loecker (2011); De Loecker and Scott (2024). De Loecker and Scott (2024) also discuss how production-side markup estimates could provide an alternative avenue for testing conduct.

clusions about firm behavior. So, while the IV approach is widely applicable, it also invites an important question: which features of conduct models actually drive instruments' relevance?

In this paper, we develop a framework that answers this question. We make three main contributions. First, we show that instruments' "reduced-form" causal effects on equilibrium prices are the fundamental economic determinant of relevance for IV-based conduct testing. This result provides a formal bridge between two historically separate empirical traditions: Sumner (1981)'s influential regression-based approach and the modern IV framework of Berry and Haile (2014). Sumner used reduced-form estimates of a cost shifter's causal effect on prices – i.e., its "pass-through" – to infer conduct, but this method is valid only for a narrow class of models requiring specific cost shifters (Bulow and Pfleiderer, 1983; MacKay et al., 2014).[2] We show the IV approach preserves Sumner (1981)'s core economic intuition – that conduct is identified through causal effects on prices – while enjoying much broader validity.

Second, we characterize exactly how each standard instrument distinguishes competing conduct models. A central theme emerges: instruments' causal effects on prices depend crucially on the cost pass-through matrix, which captures how all firms' equilibrium prices respond to changes in each firm's marginal cost. As cost pass-through plays a crucial role in determining instrument relevance, the IV approach strictly generalizes Sumner's reduced-form approach. However, different instruments rely on distinct features of the pass-through matrix to falsify incorrect models: we say that an instrument "targets" those features. Rival cost shifters target ratios of off-diagonal to diagonal elements of the pass-through matrix. Product characteristics can target the entire pass-through matrix when own-product variation is included. Tax instruments have unique properties: unit taxes target row sums of pass-through matrices, while ad valorem taxes additionally capture markup levels. These characterizations explain why particular instrument choices may lack power for specific model comparisons: they may target pass-through features that do not differ across the candidate models.

Third, we connect instrument selection to specific counterfactual objectives. When researchers ultimately care about tax policy, firm conduct is relevant insofar as it informs the tax's causal effect on prices. Our analysis suggests an important implication: historical variation in the relevant tax rate provides the ideal instrument, as it directly targets the pass-through features that determine the Laffer curve. When such variation is unavailable, our framework reveals the relevant trade-offs among alternative instruments and can even provide an avenue toward replicating the first-best instrument. This insight extends beyond taxation to any counterfactual exercise: researchers should select instruments that target the specific conduct features relevant to their policy question.

The practical implications of our results are immediate: (i) researchers should select in-

[2] Specifically, Sumner (1981) used a unit tax as the cost shifter.

struments based on their ultimate research objective—e.g., when interested in tax policy, historical tax variation provides ideal instruments; (ii) researchers should consider how candidate models differ in their implied pass-through patterns, and choose instruments that target the features exhibiting the largest differences; and (iii) researchers performing conduct inference should validate that their demand specification allows sufficient flexibility in pass-through patterns. Recently, Birchall, Mohapatra, and Verboven (2024) and Miravete, Seim, and Thurk (2026) show that standard demand specifications (including common implementations of the random coefficients logit model) can artificially restrict pass-through patterns, potentially biasing inference. Our work reveals why this matters for conduct testing: differences in pass-through matrices are what allow instruments to distinguish between conduct models. When demand curvature is restricted, conduct tests may systematically favor certain models regardless of the truth. Researchers should therefore either use demand models that allow flexible curvature when data permit, or conduct robustness checks across curvature specifications.

We illustrate all three practical implications in an application to Washington's marijuana market, computing the (counterfactual) Laffer curve for a state-level ad valorem tax. The Laffer curve in recreational marijuana markets is of substantial policy interest (Hollenbeck and Uetake, 2021), and firm conduct is one of its key determinants (Weyl and Fabinger, 2013; Miravete et al., 2018; Hollenbeck and Uetake, 2021). Following legalization in Washington, it is unclear whether retailers behave strategically or follow simple pricing rules (Escudero, 2018; Hollenbeck, Hristakeva, and Uetake, 2026). We test sophisticated Bertrand-Nash pricing (hereafter Bertrand) against a simple rule-of-thumb where price equals twice marginal cost, known as *Keystone* pricing (Escudero, 2018). Following our framework's guidance, we use ad valorem tax variation, which our analysis identifies as particularly powerful for this counterfactual, and perform robustness checks across demand curvature specifications.

Our application yields a striking finding: during our sample period (2014–2017), retailers' pricing is better explained by the Keystone rule-of-thumb, "set price at twice marginal cost," than by profit-maximizing Bertrand competition. This has economically significant implications for the Laffer curve and represents a substantial departure from the standard modeling assumption in industrial organization. We thus also contribute to an emerging literature that documents large departures from standard profit-maximizing behavior by firms. DellaVigna and Gentzkow (2019) and Adams and Williams (2019) document uniformity in prices that suggests firms are less responsive to changes in demand conditions than economists often assume. Butters, Sacks, and Seo (2022) provide evidence that pass-through of cost changes is much larger than pass-through of demand shifts in consumer packaged goods. Our finding that Keystone pricing better fits the data than Bertrand in Washington's marijuana market

is consistent with this broader pattern.

Our framework complements recent inferential advances by Backus et al. (2021) and Duarte et al. (2024), who develop the econometric properties of IV-based conduct tests. While their focus is on statistical properties in finite samples, we provide an economic framework for understanding *when and why* instruments are relevant, enabling sharp inference on conduct. This enables researchers to select instruments ex-ante based on economic reasoning by asking what features of pass-through their instruments target, rather than relying solely on ex-post diagnostics that may not reveal why instruments are weak. Our framework also motivates extensions and guides instrument selection in related settings; for instance, Duarte, Magnolfi, Quint, Sølvsten, and Sullivan (2026) rely on our results to extend the conduct testing toolkit to environments with non-constant marginal costs.

Finally, we contribute to a long line of literature on optimal tax policy, where we add to a growing body of work that performs inference on conduct to better inform tax policy (Kroft, Laliberté, Leal-Vizcaíno, and Notowidigdo, 2023; Brugués and De Simone, 2026). This practice is a recent development; much of the existing literature follows the seminal contribution of Ramsey (1927) in assuming that firms are highly competitive and set prices equal to marginal cost (Auerbach, 1985, 2002; Slemrod, 1990). That assumption can lead to misleading conclusions when the true model of conduct differs from marginal cost pricing (e.g. Suits and Musgrave, 1953; Anderson, de Palma, and Kreider, 2001; Jaffe and Weyl, 2013; Miravete et al., 2018; Hollenbeck and Uetake, 2021). Inference on firm conduct thus improves the accuracy of the researcher's findings. We also provide guidance on how to choose instruments that target the features of conduct most relevant to tax policy.

The paper proceeds as follows. Section 2 establishes our main theoretical result linking falsification to causal effects on prices. Section 3 characterizes which features of pass-through each instrument type targets, deriving conditions under which rival cost shifters, product characteristics, and taxes can distinguish competing models. Section 4 discusses implications for empirical practice, including demand estimation, counterfactual targeting, and implementation. Section 5 applies our framework to Washington's marijuana market. Section 6 concludes. Proofs of all lemmas and propositions appear in Appendix A.

2 Inference on Firm Conduct: Falsification and the Role of Pass-Through

2.1 Setup and Candidate Models

We consider the falsification of models of firm conduct using data across many markets.[3] A set of multi-product firms compete in each market t; for simplicity, we assume the same set of J products is sold in every market, although their characteristics may differ across markets. For each product and market combination (j, t), the researcher observes price p_{jt}, market share s_{jt}, a vector of product characteristics x_{jt}, and a vector of cost shifters w_{jt} that affects the product's marginal cost. For any variable a_{jt}, let a_t denote the vector of values for all products j in market t. We assume that, for all markets t, the demand system is $s_t = \mathscr{s}(p_t, x_t, \xi_t)$, where ξ_t is a vector of demand shocks that are unobserved but can be inferred by the researcher. To focus on the supply side, we assume that the demand system is already known to the researcher. That is, we proceed as if $\mathscr{s}(\cdot)$ is known and ξ_t is effectively observed.[4] We normalize market size to 1, so that quantity q_{jt} and market share s_{jt} can be used interchangeably.

Consider, as an example, a researcher who wants to know the extent to which observed prices can be attributed to markups versus costs (e.g., Atalay et al., 2025; Grieco et al., 2023; Döpper et al., 2025). The researcher knows that observed prices can be decomposed into the sum of markups and marginal costs: $p_{jt} = \Delta_{jt} + c_{jt}$. If the researcher directly observed marginal cost, then they could immediately compute the markup as $\Delta_{jt} = p_{jt} - c_{jt}$. However, neither Δ_{jt} nor c_{jt} is observed in the data, so direct inference on markups is infeasible.

Standard practice in this scenario is to impose additional structure on the data-generating process, which then enables the researcher to infer the true markups from the observed data and the already-identified demand system. To that end, we assume the data in each market t are generated by equilibrium play in some true model of firm behavior, indexed as $m = 0$.

Assumption 1. *(Equilibrium Markups and Prices)* In the true model ($m = 0$), equilibrium prices are determined as the solution to a system of first-order conditions,

$$p = \Delta_0 (p, x_t, \xi_t) + c_{0t}, \tag{1}$$

[3] Falsification encompasses cases of nested and non-nested models. We focus on the latter. For the former, researchers can pursue identification of conduct; we discuss the usefulness of our framework for this approach in Appendix C.

[4] In standard demand analysis, $\mathscr{s}(\cdot)$ is assumed to be invertible in its last argument, so $\xi_t = \mathscr{s}^{-1}(p_t, x_t, s_t)$. See Berry and Haile (2021) for a review of demand estimation methods.

where $\Delta_0(\cdot)$ is the true "markup function."[5] Furthermore, the observed prices are equilibrium prices generated by the true model, or $p_t = p_0(c_{0t}, x_t, \xi_t)$, where $p_0(\cdot)$ is the implicit function defined by Equation (1).

Under Assumption 1, the true markups $p_t - c_{0t}$ can be expressed as a function of only observed variables and the demand shocks, (p_t, x_t, ξ_t), which excludes the true marginal cost c_{0t}. It is satisfied by most static equilibrium models derived from economic theory, including models of joint profit maximization, Bertrand competition, Cournot-Nash competition (hereafter Cournot), common ownership (Backus et al., 2021), profit weighting (Miller and Weinberg, 2017), and double marginalization (Villas-Boas, 2007).[6] It also accommodates a "rule-of-thumb" where the price is a fixed multiple of marginal cost. Example 1 below provides explicit markup function formulas for five of these models, which are used extensively throughout the paper.

Example 1: Here, we explicitly characterize the markup function for marginal cost pricing, joint profit maximization (i.e., perfect collusion), Bertrand pricing, Cournot quantity setting, and "rule-of-thumb" pricing.

When firms truly price at marginal cost, observed prices satisfy $p_t = c_{0t}$, so the markup function is $\Delta_{MC}(p_t, x_t, \xi_t) \equiv 0$. At the other extreme of competition, joint profit maximization satisfies

$$p_t = \underbrace{-\left[\frac{\partial \mathscr{s}(p_t, x_t, \xi_t)}{\partial p}\right]^{-1} \mathscr{s}(p_t, x_t, \xi_t)}_{\equiv \Delta_{JP}(p_t, x_t, \xi_t)} \quad + \quad c_{0t},$$

where the researcher has already identified the market share function $\mathscr{s}(\cdot)$ and its derivatives.

The markup functions for Bertrand pricing and Cournot quantity setting take a similar form: $\Delta_B(p_t, x_t, \xi_t) \equiv -\left[\Omega \odot \frac{\partial \mathscr{s}(p_t, x_t, \xi_t)}{\partial p}\right]^{-1} \mathscr{s}(p_t, x_t, \xi_t)$ for Bertrand and $\Delta_C(p_t, x_t, \xi_t) \equiv -\left[\Omega \odot \left[\frac{\partial \mathscr{s}(p_t, x_t, \xi_t)}{\partial p}\right]^{-1}\right] \mathscr{s}(p_t, x_t, \xi_t)$ for Cournot, where Ω is the ownership matrix and $\odot$ denotes the Hadamard, or entrywise, product.[7]

When the true model is "rule-of-thumb" pricing, firms set price as a multiple of marginal cost: $p_t = \phi c_{0t}$ for scalar multiple ϕ. Some algebra reveals that this is equivalent to the

[5] Following Berry and Haile (2014), one could equivalently write markups as functions of market shares and demand derivatives. Since we treat the demand system as known, these are themselves known functions of (p_t, x_t, ξ_t). We adopt this more compact notation because our analysis requires tracking how markups respond to changes in prices and exogenous variables.

[6] Although Assumption 1 is widely-applicable, we note that it is still a substantive assumption on conduct. For example, Assumption 1 can fail when the mapping from marginal costs to prices is non-invertible; e.g., when a firm prices a product at \$50 whenever marginal costs are between \$30 and \$40. In that case, markup cannot be computed without directly including the marginal cost c_{0t} itself.

[7] $\Omega\{j, k\}$ is equal to one if products j and k are owned by the same firm, and zero otherwise; $C = A \odot B$ if $c_{ij} = a_{ij} b_{ij}$ for every (i, j).

condition $p_t = \frac{\phi-1}{\phi} p_t + c_{0t}$, so that $\Delta_\phi(p_t, x_t, \xi_t) \equiv \frac{\phi-1}{\phi} p_t$. A notable special case of rule-of-thumb pricing is the "Keystone" pricing model (Escudero, 2018), where firms set prices at twice marginal cost ($\phi = 2$) and therefore $\Delta_K(p_t, x_t, \xi_t) \equiv \frac{1}{2} p_t$. Marginal cost pricing – already discussed above – is also a special case of rule-of-thumb with $\phi = 1$. •

If the researcher knows the true model of conduct *a priori* then Assumption 1 enables the researcher to infer the true markups by computing $\Delta_{0t} = \Delta_0(p_t, x_t, \xi_t)$; this strategy is used by Atalay et al. (2025); Grieco et al. (2023); Döpper et al. (2025), who all assume Bertrand price competition. We focus on settings where the researcher is uncertain about the form of firm conduct, and instead considers a menu of theoretically plausible models, asking which are falsified by the data. For each model m, the researcher computes its implied markups, $\Delta_{mt} \equiv \Delta_m(p_t, x_t, \xi_t)$. Given the markups, model-implied marginal costs c_{mt} are calculated as those that rationalize the observed prices: $c_{mt} = p_t - \Delta_{mt}$. When only one model on the menu is not falsified, the researcher then treats it as the true model, and treats Δ_{mt} as the true markups.

Aside from recovering accurate markups, there are several other reasons conduct inference is useful to economists. First, researchers may investigate firm conduct for its own sake. For example, they may wonder whether firms coordinate on prices (Miller and Weinberg, 2017; Miller, Sheu, and Weinberg, 2021; Calder-Wang and Kim, 2024) or whether common ownership incentives affect pricing decisions (Backus et al., 2021). Second, researchers might be interested in using the structural equilibrium model to perform counterfactual policy analysis. For example, the researcher may want to compute the Laffer curve for a market, which requires counterfactual simulations of market outcomes over potential tax rates. Recent empirical research has shown that firm conduct is a substantial determinant of the Laffer curve (Miravete et al., 2018; Hollenbeck and Uetake, 2021), so it is important that the researcher be able to determine firm conduct. Regardless of the researcher's motivation for learning conduct, the equilibrium restriction in Assumption 1 is crucial for conduct inference.

2.2 Falsification via Exclusion Restrictions

The seminal work of Bresnahan (1982) and Lau (1982) establishes that conduct can be learned by leveraging exogenous variation in market conditions. Berry and Haile (2014) generalize this intuition to differentiated products markets, showing that a broad class of exogenous variation, combined with restrictions on the cost function, generates falsifiable restrictions that can discriminate between conduct models. We specialize their framework to the following setting.

Assumption 2. *(Constant Marginal Cost)* Marginal costs are constant in quantities and take the form $c_{0jt} = \bar{c}_{0j}(\mathrm{w}_{jt}) + \omega_{0jt}$ for all j, where ω_{0jt} is an unobserved cost shock. Fur-

thermore, $\bar{c}_{0j}$ is differentiable.

Assumption 3. *(Instrument Exogeneity)* There is a vector of K instruments, $z_{jt} \subseteq \{x_{\ell t}, \mathrm{w}_{\ell t}\}_{\ell=1}^{J}$ such that (i) $E[\omega_{0jt} \mid \mathrm{w}_{jt}, z_{jt}] = 0$ *a.s.*;[8] (ii) $\frac{\partial \bar{c}_{0j}}{\partial z_{jt}}$ is known *a priori*; and (iii) the researcher only considers cost functions such that $\frac{\partial \bar{c}_{mj}}{\partial z_{jt}} = \frac{\partial \bar{c}_{0j}}{\partial z_{jt}}$ *a.s.*

Assumption 2 imposes constant marginal cost, ruling out economies of scale. This assumption has been heavily utilized in the recent literature on conduct inference (Backus et al., 2021; Duarte et al., 2024), and it is widely-used in the broader empirical IO literature (e.g., Miller and Weinberg (2017); Miravete et al. (2018)).[9] We allow for overlap between x_{jt} and w_{jt}, so that product characteristics can potentially impact marginal costs. For our running examples and our empirical application, we will further assume that marginal costs are linear in cost shifters, $\bar{c}_{0j}(\mathrm{w}_{jt}) = \mathrm{w}_{jt}'\gamma_0$, as is standard in the empirical literature; but we make no such assumption for our general falsification results.

Assumption 3 gives the key falsifiable restriction, requiring exogenous instruments to have known effects on marginal cost. Previous work has focused on instruments that are *excluded* from marginal cost (Berry and Haile, 2014; Backus et al., 2021; Duarte et al., 2024); for example, rival products' marginal cost shifters or characteristics. Exclusion is a special case of Assumption 3, where $\frac{\partial \bar{c}_{0j}}{\partial z_{jt}} = 0$. Assumption 3 relaxes exclusion and can accommodate a wider variety of instruments. For example, suppose that firms are retailers and that z_{jt} is the wholesale price for product j. Then $\frac{\partial \bar{c}_{0j}}{\partial z_{jt}} = 1$, because marginal cost increases one-for-one with wholesale price. A wholesale price therefore satisfies Assumption 3 even though it is not excluded from the marginal cost function.

When true costs and markups are unobserved, the exclusion restriction summarizes what we know about the true model: $E[\omega_{0jt} \mid \mathrm{w}_{jt}, z_{jt}] = 0$ holds for the true cost function, and $\frac{\partial \bar{c}_{0j}}{\partial z_{jt}}$ is already known. For a candidate model m and candidate cost function $\bar{c}_{mj}$ such that $\frac{\partial \bar{c}_{mj}}{\partial z_{jt}} = \frac{\partial \bar{c}_{0j}}{\partial z_{jt}}$, we can use Equation (1) to define $\omega_{mjt} \equiv p_{jt} - \Delta_{mjt} - \bar{c}_{mj}(\mathrm{w}_{jt})$, analogous to ω_{0jt}. That is, ω_{mt} is the vector of cost shocks that rationalizes the observed prices, given Δ_{mt} and $\bar{c}_{mt}$. Thus, $E[\omega_{mjt} \mid \mathrm{w}_{jt}, z_{jt}]$ provides a natural way to measure a model's fit. When $E[\omega_{mjt} \mid \mathrm{w}_{jt}, z_{jt}] = 0$,[10] the model has the same fit as the true model, so it is not falsified by the instruments z_{jt}. Conversely, when there are no suitable cost functions $\{\bar{c}_{mj}(\mathrm{w}_{jt})\}_{j=1}^{J}$ satisfying this falsifiable restriction almost surely over the values of w_{jt} and z_{jt}, the conduct model m is *falsified by the instruments* z_{jt}.

[8] Here, as in the rest of the paper, the expectation is taken over realizations of unobservables, and *almost surely (a.s.)* refers to realizations of the exogenous observables $(\mathrm{w}_{jt}, z_{jt})$.

[9] In follow-on work, Duarte et al. (2026) show that our framework can be extended to accommodate scale economies. Moreover, the falsification results we obtain under the constant cost assumption provide necessary conditions for falsification under non-constant costs.

[10] This expression is the analogue of the condition in Theorem 9 of Berry and Haile (2014).

Valid instruments are exogenous, but they may not be relevant for falsification. In that case, the data will fail to falsify an untrue model. However, determining instrument relevance from the falsifiable restriction is not obvious. To gain insight into the economic determinants of instrument relevance, it is useful to follow Backus et al. (2021) and rewrite the falsifiable restriction in terms of markups, an economic implication of the model. As prices in the data are generated from the true model in Equation (1), we can express the model specific unobservable ω_{mjt} as $\omega_{mjt} = \Delta_{0jt} - \Delta_{mjt} + \bar{c}_{0j}(\mathrm{w}_{jt}) - \bar{c}_{mj}(\mathrm{w}_{jt}) + \omega_{0jt}$. From this, we can restate the falsifiable restriction as follows.

Lemma 1. *Under Assumptions 1-2, model m is falsified by instruments z_{jt} if and only if for some j there exists no function $\bar{c}_{mj}(\mathrm{w}_{jt})$ such that*

$$E[\Delta_{0jt} - \Delta_{mjt} \mid \mathrm{w}_{jt}, z_{jt}] = \bar{c}_{mj}(\mathrm{w}_{jt}) - \bar{c}_{0j}(\mathrm{w}_{jt}) \qquad a.s.$$

Recent innovations in the literature have focused on *statistical* implications of this condition (Backus et al., 2021; Duarte et al., 2024). We will use it to determine the underlying *economic* forces that drive instrument relevance. While Assumption 3 places restrictions on some derivatives of the marginal cost function, its location is flexible. So, as discussed by Backus et al. (2021), falsification requires *changes* in the values of the instruments to be predictive of changes in the expected difference in markups; we need to consider how the expected markup difference moves with respect to changes in the instrument vector. Thus, it is useful to restate Lemma 1 in terms of the marginal impacts of the instruments on markups. To do this, we will assume markups move continuously as instruments change:[11]

Assumption 4. *(Continuous Markups)* For any model m, for any j, $E[\Delta_{mjt} \mid \mathrm{w}_{jt}, z_{jt}]$ is absolutely continuous in z_{jt}.

Lemma 2. *Under Assumptions 1-4, model m is falsified by instruments z_{jt} if and only if for some j and k,*

$$E\left[\frac{\mathrm{d}\Delta_{0jt}}{\mathrm{d}z_{jt}} - \frac{\mathrm{d}\Delta_{mjt}}{\mathrm{d}z_{jt}} \mid \mathrm{w}_{jt}, z_{jt}\right] \neq 0 \qquad w.p.p.$$

Lemma 2 tells us that falsification requires instruments to differentially shift markups across models, a point emphasized in Berry and Haile (2021). But what economic features of a conduct model determine whether this occurs? We turn to this in the next section.

[11] Assumption 4 holds for all models in this paper, and for all standard models.

2.3 Pass-Through and Main Result

To build intuition for what determines falsification, it is useful to state the condition in Lemma 2 in terms of the causal effect of the instrument on prices. This, in turn, will depend on the cost pass-through matrix, which we therefore define first.

Under Assumptions 1 and 2, the true cost pass-through matrix is defined as

$$P_{0t} \quad = \quad \left(I - \frac{\partial \Delta_0\left(p_t, x_t, \xi_t\right)}{\partial p}\right)^{-1} \quad \equiv \quad \frac{\mathrm{d}p_0\left(c_{0t}, x_t, \xi_t\right)}{\mathrm{d}c},$$

which follows from applying the implicit function theorem to Equation (1). Similarly, the pass-through matrix implied by model m is $P_{mt} = \left(I - \frac{\partial \Delta_m(p_t, x_t, \xi_t)}{\partial p}\right)^{-1}$. We assume that pass-through matrices are invertible:[12]

Assumption 5. *(Invertibility of Pass-Throughs)* For any model m, in any market t, the pass-through matrix P_{mt} has full rank.

Intuitively, this simply requires different cost vectors to produce different equilibrium prices. We provide several examples of invertible pass-through matrices in Section 3.

Next, we define the instrument's *causal effect on equilibrium prices.* With $z_{jt} \subseteq \{x_{\ell t}, \mathrm{w}_{\ell t}\}_{\ell=1}^{J}$, implicit differentiation of (1) yields[13]

$$\frac{\mathrm{d}p_{0t}}{\mathrm{d}z_{jt}} = P_{0t}\left(\frac{\partial \Delta_{0t}}{\partial z_{jt}} + \frac{\partial \bar{c}_{0t}}{\partial z_{jt}}\right). \tag{2}$$

For model m with Δ_{mt}, the model-implied causal effect of the instrument on equilibrium prices also depends on the researcher's choice of marginal cost function $\bar{c}_{mt}(\cdot)$. Given the marginal cost function, the model-implied causal effect is $\frac{\mathrm{d}p_{mt}}{\mathrm{d}z_{jt}} = P_{mt}\left(\frac{\partial \Delta_{mt}}{\partial z_{jt}} + \frac{\partial \bar{c}_{mt}}{\partial z_{jt}}\right)$. We note that only the term $\frac{\partial \bar{c}_{mt}}{\partial z_{jt}}$ depends on the marginal cost function; all other terms depend on the markup function and its derivatives. Using these formulas for the causal effects on prices, along with the identity $\frac{\partial \Delta_{0t}}{\partial p_t} - \frac{\partial \Delta_{mt}}{\partial p_t} = P_{mt}^{-1} - P_{0t}^{-1}$, we obtain a useful alternative representation of the difference in markup derivatives in Lemma 2, which leads to our main result. Here, $(A)_j$ denotes the jth row of a vector or matrix.

[12] This assumption has economic content, as it requires that each product has non-zero pass-through for at least one cost (either own or rival) in the market. Moreover, pass-through vectors for each product cannot be linear combinations of those of other products, meaning that costs must affect different products in a distinct way. This is satisfied in most models, as the pass-through of own cost that is measured by the main diagonal of P_{mt} is typically different than the pass-through of rival costs.

[13] With $c_{0t} \equiv \bar{c}_{0t}(\mathrm{w}_t) + \omega_{0t}$, the price p_{0t} is the solution to an implicit function $p = \Delta_0(p, x_t, \xi_t) + \bar{c}_0(w_t) + \omega_{0t}$. The Implicit Function Theorem then yields $\frac{\mathrm{d}p_{0t}}{\mathrm{d}z_t} = \left(I - \frac{\partial \Delta_{0t}}{\partial p_t}\right)^{-1}\left(\frac{\partial \Delta_{0t}}{\partial z_t} + \frac{\partial \bar{c}_{0t}}{\partial z_{jt}}\right) \equiv P_{0t}\left(\frac{\partial \Delta_{0t}}{\partial z_t} + \frac{\partial \bar{c}_{0t}}{\partial z_{jt}}\right)$.

Lemma 3. *Under Assumptions 1-5, given model m's implied markup function $\Delta_m(\cdot)$, the following equivalence holds for* any *candidate marginal cost function:*

$$\frac{\mathrm{d}\Delta_{0jt}}{\mathrm{d}z_{jt}} - \frac{\mathrm{d}\Delta_{mjt}}{\mathrm{d}z_{jt}} = \left(P_{mt}^{-1}\right)_j \left(\frac{\mathrm{d}p_{0t}}{\mathrm{d}z_{jt}} - \frac{\mathrm{d}p_{mt}}{\mathrm{d}z_{jt}}\right).$$

Proposition 1. *Under Assumptions 1-5, a model m is falsified by instruments $z_{jt} \subseteq \{x_{\ell t}, \mathrm{w}_{\ell t}\}_{\ell=1}^{J}$ if and only if every admissible marginal cost function $\bar{c}_m(\cdot)$ is such that*

$$E\left[\left(P_{mt}^{-1}\right)_j \left(\frac{\mathrm{d}p_{0t}}{\mathrm{d}z_{jt}} - \frac{\mathrm{d}p_{mt}}{\mathrm{d}z_{jt}}\right) \mid \mathrm{w}_{jt}, z_{jt}\right] \neq 0 \qquad w.p.p. \tag{3}$$

Proposition 1 highlights the fundamental role of instruments' "reduced-form" causal effects on prices, $\frac{\mathrm{d}p}{\mathrm{d}z}$, when performing conduct inference: falsification requires the model-implied causal effect – determined by both the conduct model and the marginal cost function – to differ from the true causal effect.[14] Proposition 1 therefore connects to a literature that uses reduced-form approaches to learn conduct, beginning with the path-breaking work of Sumner (1981). Sumner directly estimated the causal effect of an excise tax on cigarette prices using reduced-form regressions, and a model was falsified if its implied causal effect did not match the regression coefficient. However, Sumner's approach requires the causal effect to be a constant structural parameter, limiting its validity to a narrow class of models (Bulow and Pfleiderer, 1983; MacKay et al., 2014). Proposition 1 shows that the core intuition – conduct is learned from reduced-form causal effects on prices – extends to the modern IV framework of Berry and Haile (2014), which accommodates models where pass-through varies with market conditions. We delve even deeper into this connection in Appendix B, showing that the instrumental variable approach can be viewed as a strict generalization of the regression-based approach.

While Proposition 1 is stated in terms of causal effects on prices, these objects may not immediately reveal which economic features of a conduct model actually drive falsification. In fact, the causal effects estimated in Sumner (1981) were reduced-form measures of cost pass-through rates because costs increase one-to-one with a unit tax, so that falsification amounted to comparing the pass-through implied by competing conduct models to the pass-through estimated from the data. We can show that the falsifiable restriction in Proposition 1 also relies on a comparison of cost pass-through. Using Equation (2) and Assumption 3, we

[14] For instrument z_{jt}, Assumption 3 only restricts the partial derivative of the cost function for product j. The condition in Proposition 1 must therefore hold for all candidate cost functions for the other products.

can rewrite the difference in causal effects as:[15]

$$\frac{\mathrm{d}p_{0t}}{\mathrm{d}z_{jt}} - \frac{\mathrm{d}p_{mt}}{\mathrm{d}z_{jt}} = P_{mt}\left[\left(\frac{\partial\Delta_{0t}}{\partial z_{jt}} - \frac{\partial\Delta_{mt}}{\partial z_{jt}}\right) + \left(P_{mt}^{-1} - P_{0t}^{-1}\right)\frac{\mathrm{d}p_{0t}}{\mathrm{d}z_{jt}}\right].$$

The first term inside the brackets captures direct effects, arising when the instrument enters the markup function differently across models, as can occur with product characteristic instruments. The second term depends on differences in inverse pass-through matrices: when $P_{mt} \neq P_{0t}$ and the instrument moves prices under the true model, this term contributes to falsification.

For instruments that do not directly enter markups, such as cost shifters and taxes, the first term vanishes, and falsification is driven entirely by differences in pass-through. Even when direct effects are present, differences in pass-through are generically sufficient for falsification, since the two terms inside the brackets would have to exactly offset to prevent it. In this sense, differences in the true cost pass-through matrix and the one implied by model m play a crucial role in determining instrument relevance. We characterize the specific features of pass-through targeted by each instrument type in Section 3.

3 Evaluating Instrument Relevance

In this section, we evaluate the relevance of instruments commonly used in the literature. Given the results above, we focus on the features of the conduct model itself – i.e., the implied markup function and pass-through matrix – that drive falsification with various instruments.

We start with standard cost shifter instruments with no direct effects on markups, and instruments based on product characteristics that may have direct effects on markups. We then augment Proposition 1 for settings with taxes, and explore the relevance of tax instruments. We show that different instruments target different features of the conduct model.

We then show how our analysis can help the researcher select instruments when their ultimate goal is to perform counterfactual analysis. Conduct has a first-order impact on counterfactual analysis, but only some features of the conduct model may be important for a given counterfactual. Our analysis enables researchers to select "counterfactual-relevant" instruments that target those specific features of conduct.

Before proceeding, we make two additional remarks. First, we note that the falsification condition (3) in Proposition 1 applies to conditional *expectations*, so it is still possible that a

[15] From Equation (2), $\frac{\mathrm{d}p_{mt}}{\mathrm{d}z_{jt}} = P_{mt}\left(\frac{\partial\Delta_{mt}}{\partial z_{jt}} + \frac{\partial\bar{c}_{mt}}{\partial z_{jt}}\right)$ for any model m. Taking the difference between the true and candidate models, and using $\frac{\partial\bar{c}_{mt}}{\partial z_{jt}} = \frac{\partial\bar{c}_{0t}}{\partial z_{jt}}$ (Assumption 3), we obtain $\frac{\mathrm{d}dp_{0t}}{\mathrm{d}z_{jt}} - \frac{\mathrm{d}p_{mt}}{\mathrm{d}z_{jt}} = P_{0t}\frac{\partial\Delta_{0t}}{\partial z_{jt}} - P_{mt}\frac{\partial\Delta_{mt}}{\partial z_{jt}} + (P_{0t} - P_{mt})\frac{\partial\bar{c}_{0t}}{\partial z_{jt}}$. Substituting $\frac{\partial\bar{c}_{0t}}{\partial z_{jt}} = P_{0t}^{-1}\frac{\mathrm{d}p_{0t}}{\mathrm{d}z_{jt}} - \frac{\partial\Delta_{0t}}{\partial z_{jt}}$ and collecting terms yields the result.

model will not be falsified even when the term inside the expectation is non-zero w.p.p. because it can still average out to zero. However, we view this situation as an unlikely "knife's edge" case, and we generally expect falsification to hold whenever the expression inside the expectation in (3) is non-zero w.p.p.. Much of our ensuing discussion utilizes this intuition to interpret the empirical content of various instruments.

Second, we note that we present several examples throughout our analysis. Most of them focus on a simple environment described in Example 2 below.

Example 2: In this and every subsequent example, we consider a simple environment with two single-product firms and logit demand, so that market shares for $j \in \{1, 2\}$ are given by

$$s_{jt} = \frac{\exp(x_{jt}\beta - \alpha p_{jt})}{1 + \exp(x_{1t}\beta - \alpha p_{1t}) + \exp(x_{2t}\beta - \alpha p_{2t})},$$

where x_{jt} are characteristics of product j in market t and α and β are coefficients.

We calculate equilibrium markups and pass-through matrices for the five conduct models introduced in Example 1. First is marginal cost pricing, $p_{jt} = c_{0jt}$, under which markups are $\Delta_{MCjt} = 0$, and pass-through and inverse pass-through matrices are $P_{MCt} = P_{MCt}^{-1} = I$. Second is joint profit maximization, which gives markups, inverse pass-through, and pass-through matrices

$$\Delta_{JPt} = \begin{bmatrix} \frac{1}{\alpha(1-s_{0t})} \\ \frac{1}{\alpha(1-s_{0t})} \end{bmatrix}, \quad P_{JPt} = \begin{bmatrix} 1 - s_{1t} & -s_{2t} \\ -s_{1t} & 1 - s_{2t} \end{bmatrix}, \quad \text{and} \quad P_{JPt}^{-1} = \frac{1}{s_{0t}} \begin{bmatrix} 1 - s_{2t} & s_{2t} \\ s_{1t} & 1 - s_{1t} \end{bmatrix}$$

where $s_{0t} = 1 - s_{1t} - s_{2t}$. Third is Bertrand price competition, with

$$\Delta_{Bt} = \begin{bmatrix} \frac{1}{\alpha(1-s_{1t})} \\ \frac{1}{\alpha(1-s_{2t})} \end{bmatrix}, \quad P_{Bt} = \kappa_{Bt} \begin{bmatrix} \frac{1}{1-s_{2t}} & \frac{s_{1t}s_{2t}}{(1-s_{1t})^2} \\ \frac{s_{1t}s_{2t}}{(1-s_{2t})^2} & \frac{1}{1-s_{1t}} \end{bmatrix}, \quad \text{and} \quad P_{Bt}^{-1} = \begin{bmatrix} \frac{1}{1-s_{1t}} & -\frac{s_{1t}s_{2t}}{(1-s_{1t})^2} \\ -\frac{s_{1t}s_{2t}}{(1-s_{2t})^2} & \frac{1}{1-s_{2t}} \end{bmatrix}$$

where $\kappa_{Bt} = s_{0t}^{-1}(1 - s_{1t})^2(1 - s_{2t})^2$. Fourth is the differentiated-products version of Cournot competition in quantities, where we have

$$\Delta_{Ct} = \begin{bmatrix} \frac{1-s_{2t}}{\alpha s_{0t}} \\ \frac{1-s_{1t}}{\alpha s_{0t}} \end{bmatrix}, \quad P_{Ct} = \begin{bmatrix} \frac{s_{0t}}{1-s_{2t}} & 0 \\ 0 & \frac{s_{0t}}{1-s_{1t}} \end{bmatrix}, \quad \text{and} \quad P_{Ct}^{-1} = \begin{bmatrix} \frac{1-s_{2t}}{s_{0t}} & 0 \\ 0 & \frac{1-s_{1t}}{s_{0t}} \end{bmatrix}.$$

Last is the rule-of-thumb model, where firms set price at ϕ times marginal cost. There, we have markups $\Delta_{\phi t} = \frac{\phi-1}{\phi} p_t$, cost pass-through matrix $P_{\phi t} = \phi I$, and inverse pass-through matrix $P_{\phi t}^{-1} = \frac{1}{\phi} I$. In the special case of the Keystone model, prices are set at twice marginal costs ($\phi = 2$) and we have $\Delta_{Kt} = \frac{1}{2} p_t$, $P_{Kt} = 2I$, and $P_{Kt}^{-1} = \frac{1}{2} I$.

We see that the pass-through matrix is invertible for all five conduct models (Assumption 5). Also in this simple environment, the Cournot, Keystone, and marginal cost pricing models all have zero pass-through of rival's costs (i.e., zero off-diagonal elements for P_{mt}). This will be important in future examples. •

3.1 Cost Shifter Instruments

Here, we consider marginal cost shifters – that have no direct effect on demand or markups – as instruments. Let $\mathrm{w}_t = (\mathrm{w}_{1t}, \ldots, \mathrm{w}_{Jt})$ be a vector of scalar-valued, product-specific marginal cost shifters. To simplify the exposition, we assume that the cost function takes a linear form, so that $\bar{c}_{0jt} = \gamma_0^{\text{const}} + \gamma_0 \mathrm{w}_{jt}$.[16] The cross-derivatives are known *a priori* because they are zero due to exclusion: $\frac{\partial \bar{c}_{0jt}}{\partial \mathrm{w}_{\ell t}} = 0$ for $j \neq \ell$. However, the own derivative $\frac{\partial \bar{c}_{0jt}}{\partial \mathrm{w}_{jt}} = \gamma_0$ is only known when γ_0 is known.

Before proceeding with the analysis, we introduce a couple of useful identities. First, the effect of product ℓ's cost shifter on the vector of marginal costs is $\frac{\partial \bar{c}_{0t}}{\partial \mathrm{w}_{\ell t}} = \gamma_0 e_\ell$, where e_ℓ is the ℓ-th standard basis vector. Cost shifters have no direct effects on markups, so (2) reduces to

$$\frac{\mathrm{d}p_{0t}}{\mathrm{d}\mathrm{w}_{\ell t}} = \gamma_0 P_{0t} e_\ell.$$

Similarly, for model m we have $\frac{\partial \bar{c}_{mt}}{\partial \mathrm{w}_{\ell t}} = \gamma_m e_\ell$ and $\frac{\mathrm{d}p_{mt}}{\mathrm{d}\mathrm{w}_{\ell t}} = \gamma_m P_{mt} e_\ell$.

Rival cost shifters (unknown γ_0): When γ_0 is unknown, only rival product's cost shifters can be used as instruments, since only $\frac{\partial \bar{c}_{0t}}{\partial \mathrm{w}_{\ell t}} = 0$ for $j \neq \ell$ is known *a priori*. Setting $z_{jt} = \mathrm{w}_{\ell t}$ for $j \neq \ell$, and noting that $\left(P_{mt}^{-1}\right)_j = e_j' P_{mt}^{-1}$, the falsification condition in Proposition 1 reduces to

$$E\left[e_j' \left(\gamma_0 P_{mt}^{-1} P_{0t} - \gamma_m I\right) e_\ell \mid \mathrm{w}_{jt}, z_{jt}\right] \neq 0 \quad w.p.p.$$

for any $j \neq \ell$.

For any matrix A we have $e_j' A e_\ell = (A)_{j\ell}$. So, falsification requires

$$\text{off-diag}\left\{\gamma_0 P_{mt}^{-1} P_{0t} - \gamma_m I\right\} \neq 0.$$

for some γ_m. This occurs if and only if $P_{mt}^{-1} P_{0t}$ is not a diagonal matrix. $P_{mt}^{-1} P_{0t}$ not being a diagonal matrix requires $P_{0t} \neq P_{mt}\Lambda$ for any diagonal matrix Λ, meaning the columns of P_{0t} cannot each be a rescaling of the corresponding column of P_{mt}; assuming the diagonal

[16] Linear cost functions are used by Duarte et al. (2024), and we use their PyRVtest Python package (Duarte, Magnolfi, Sølvsten, Sullivan, and Tarascina, 2022) in our empirical application.

terms of P_{mt} are non-zero, this requires there to be some pair of $j \neq \ell$ such that

$$\frac{(P_{mt})_{j\ell}}{(P_{mt})_{\ell\ell}} \neq \frac{(P_{0t})_{j\ell}}{(P_{0t})_{\ell\ell}}.$$

If instead this ratio is the same across the two models, then there is some γ_m that replicates the true causal effects of rival cost shifters on prices with model m, leading to lack of falsification.[17]

Remark 1. Rival cost shifter instruments target the *ratio* of the cost pass-through matrix's off-diagonal elements to their column's diagonal element; models with $(P_{mt})_{j\ell}/(P_{mt})_{\ell\ell} \neq (P_{0t})_{j\ell}/(P_{0t})_{\ell\ell}$ for some $j \neq \ell$ are falsified.

Rival cost shifter instruments distinguish conduct models through the implied pass-through matrix, but they do not target the full pass-through matrix; they only target it up to re-scaling of the columns. So, there are cases where rival cost shifters fail to falsify an untrue model. The next example provides further illustration.

Example 3: Return to the environment from Example 2. Suppose we want to falsify models of competition using variation in the cost shifter of firm 2, w_{2t}, so that $z_{1t} = \mathrm{w}_{2t}$. The key object is the ratio of the off-diagonal and the diagonal element in the second column of the pass-through matrix: $(P_{mt})_{12}/(P_{mt})_{22}$.

First, suppose the true model is Bertrand competition and we want to falsify joint profit maximization (i.e., test for perfect collusion). Using the pass-through matrices shown in Example 2, the ratio of the relevant elements of the pass-through matrix is equal to $\frac{-s_{2t}}{1-s_{2t}}$ under joint profit maximization and $\frac{s_{1t}s_{2t}}{1-s_{1t}}$ for Bertrand. These have opposite signs, so joint profit maximization is falsified.

Now, suppose the true model is Cournot competition. The Cournot pass-through matrix is diagonal (Example 2), so the ratio of rival to own pass-through is zero. Joint profit maximization is falsified because the ratio is strictly negative under that model; Bertrand is also falsified because the ratio is strictly positive.

However, if instead we wished to falsify either the Keystone model or marginal cost pricing, we would fail. Both of these models, like Cournot, have diagonal pass-through matrices: each firm's equilibrium price responds only to its own cost, not to rivals' costs. So the ratios are zero under all three models, and rival cost shifters cannot distinguish between them. •

[17] Suppose a shifter of cost ℓ moves, resulting in changes to prices p_ℓ and (potentially) p_j. The change in p_ℓ is equal to $\frac{\mathrm{d}p_{0\ell t}}{\mathrm{d}c_{\ell t}}\frac{\partial \bar{c}_{0\ell}}{\partial \mathrm{w}_{\ell t}}$; the change in p_j is equal to $\frac{\mathrm{d}p_{0jt}}{\mathrm{d}c_{\ell t}}\frac{\partial \bar{c}_{0\ell}}{\partial \mathrm{w}_{\ell t}}$. The effect of the cost shifter $\frac{\partial \bar{c}_{0\ell}}{\partial \mathrm{w}_{\ell t}}$ is not known ex ante; but for a model where $\frac{\mathrm{d}p_{mjt}}{\mathrm{d}c_{\ell t}} \Big/ \frac{\mathrm{d}p_{m\ell t}}{\mathrm{d}c_{\ell t}}$ differs from the truth, no value of $\frac{\partial \bar{c}_{m\ell}}{\partial \mathrm{w}_{\ell t}}$ can rationalize both price changes, so ω_{mjt} cannot be mean-independent of $\mathrm{w}_{\ell t}$.

Own cost shifters (known γ_0): Now we consider the case where γ_0 is known by the researcher *a priori*. For example, $\gamma_0 = 1$ when w_t is a vector of wholesale prices. With γ_0 known, the researcher knows $\frac{\partial \bar{c}_{0jt}}{\partial \mathrm{w}_{\ell t}}$ for all (j, ℓ), including $j = \ell$. This implies that own cost shifters can be used as instruments, in addition to rival cost shifters. Additionally, the researcher restricts attention to cost functions with $\gamma_m = \gamma_0$ (Assumption 3).

The restriction in Proposition 1 now simplifies to

$$E\left[\gamma_0 e_j' \left(P_{mt}^{-1} P_{0t} - I\right) e_\ell \mid \mathrm{w}_{jt}, z_{jt}\right] \neq 0 \quad w.p.p.$$

for all (j, ℓ). So, falsification requires $P_{mt}^{-1} P_{0t} \neq I$, which occurs if and only if $P_{mt} \neq P_{0t}$; the researcher can now use the full pass-through matrix for falsification.

Remark 2. Instruments constructed with own and rival cost shifters target the *full pass-through matrix*; models with $P_{mt} \neq P_{0t}$ will be falsified.

Clearly, any model that can be falsified with rival cost shifters can also be falsified with the combination of own and rival cost shifters. Additionally, the combination can falsify models that could not be falsified by rival cost shifters alone, as we illustrate in the following example.

Example 4: We return to the environment of Example 2 and suppose the truth is Cournot competition. In that case, Example 3 showed that rival cost shifters could falsify joint profit maximization or Bertrand competition, but not Keystone or marginal cost pricing. The latter models can be falsified with the combination of own and rival cost shifters, which target the entire pass-through matrix. To see this, note that $(P_{Ct})_{11} = \frac{s_{0t}}{1-s_{2t}}$, $(P_{MCt})_{11} = 1$, and $(P_{Kt})_{11} = 2$. Because market shares vary across markets, we have $(P_{Ct})_{11} \neq (P_{MCt})_{11}$ and $(P_{Ct})_{11} \neq (P_{Kt})_{11}$ almost surely. So, Keystone and marginal cost pricing are both falsified.•

Recap: The relevance of cost side instruments depends crucially on what is known about the cost function. The full pass-through matrix can be targeted when own cost shifters' effects on marginal costs are known. When these effects are unknown, rival products' cost shifters can still be used as instruments but target more limited features of pass-through.

3.2 Product Characteristics Instruments

Product characteristics that affect demand are also candidate instruments for conduct. For simplicity, we focus on a vector of product characteristics $x_t = (x_{1t}, \ldots, x_{Jt})$ used as instruments. We allow own-product characteristics to enter the cost function, and we assume that $\bar{c}_{0jt} = \gamma_0^{\text{const}} + \gamma_0 x_{jt}$ for parsimony. Unlike cost-shifter instruments, product characteristics also have direct effects on markups. Applying the general causal effect formula in (2) with

$z_{jt} = x_{\ell t}$, the causal effect of product characteristic $x_{\ell t}$ on prices is

$$\frac{\mathrm{d}p_{0t}}{\mathrm{d}x_{\ell t}} = P_{0t}\left(\frac{\partial \Delta_{0t}}{\partial x_t} + \gamma_0 I\right) e_\ell,$$

where e_ℓ is again the ℓ-th elementary basis vector. The causal effect for model m paired with γ_m has a similar expression, with the 0 index replaced by m.

Rival product characteristics (unknown γ_0): The marginal cost coefficient γ_0 will be unknown in many applications (e.g., Backus et al., 2021), in which case only rival product characteristics can be used because their effect on product j's marginal cost is known to be zero *a priori*.

With $z_{jt} = x_{\ell t}$, the falsification condition in Proposition 1 reduces to

$$E\left[e_j'\left(P_{mt}^{-1}P_{0t}\frac{\partial \Delta_{0t}}{\partial x_t} - \frac{\partial \Delta_{mt}}{\partial x_t} + \gamma_0 P_{mt}^{-1}P_{0t} - \gamma_m I\right) e_\ell \mid \mathrm{w}_{jt}, z_{jt}\right] \neq 0 \quad w.p.p.$$

This falsification condition applies to any $j \neq \ell$, so falsification requires

$$\text{off-diag}\left\{P_{mt}^{-1}P_{0t}\frac{\partial \Delta_{0t}}{\partial x_t} - \frac{\partial \Delta_{mt}}{\partial x_t} + \gamma_0 P_{mt}^{-1}P_{0t} - \gamma_m I\right\} \neq 0$$

for some γ_m. This condition is similar to the falsification condition for rival cost shifters, which required off-diag $\left\{\gamma_0 P_{mt}^{-1}P_{0t} - \gamma_m I\right\} \neq 0$. While falsification with rival cost shifters fails whenever $P_{mt}^{-1}P_{0t}$ is diagonal, falsification can still be obtained with rival product characteristics when off-diag $\left\{P_{mt}^{-1}P_{0t}\frac{\partial \Delta_{0t}}{\partial x_t} - \frac{\partial \Delta_{mt}}{\partial x_t}\right\} \neq 0$. So, product characteristics' direct effect on markups can give them unique empirical content.

Own product characteristics (known γ_0): There are some situations where the effect of product j's characteristic on its marginal cost is known by the researcher *a priori*. For example, advertising expenditure is sometimes modeled as a product characteristic that affects demand but is excluded from the marginal cost of production (e.g., Goeree, 2008), so that $\gamma_0 = 0$. While such an exclusion restriction may be the most intuitive case, we allow for a general γ_0. With the researcher now able to use $z_{jt} = x_{\ell t}$ for any (j, ℓ) and imposing $\gamma_m = \gamma_0$ (Assumption 3), falsification now requires

$$P_{mt}^{-1}P_{0t}\frac{\partial \Delta_{0t}}{\partial x_t} - \frac{\partial \Delta_{mt}}{\partial x_t} + \gamma_0\left(P_{mt}^{-1}P_{0t} - I\right) \neq 0.$$

The term $\gamma_0\left(P_{mt}^{-1}P_{0t} - I\right)$ was the key to the falsification condition for the combination of own and rival cost shifters. Again, we see that product characteristics have distinct empirical content, since the additional term $P_{mt}^{-1}P_{0t}\frac{\partial \Delta_{0t}}{\partial x_t} - \frac{\partial \Delta_{mt}}{\partial x_t}$ can deliver falsification even when cost

shifters fail.

The role of direct effects on markups: Product characteristics' direct effects on markups play an important role in falsification, giving these instruments unique empirical content. To illustrate this, we will focus on two special cases.

First, consider a case where the model matches the true cost pass-through matrix: $P_{mt} = P_{0t}$. In that case, falsification via rival product characteristics requires the model to satisfy off-diag $\left\{\frac{\partial \Delta_{0t}}{\partial x_t} - \frac{\partial \Delta_{mt}}{\partial x_t}\right\} \neq 0$; the model is falsified when there is some pair of products $j \neq \ell$ such that the model's implied direct effect of product ℓ's characteristics on product j's price differs from the truth. With the combination of own and rival product characteristic instruments, falsification requires $\frac{\partial \Delta_{mt}}{\partial x_t} \neq \frac{\partial \Delta_{0t}}{\partial x_t}$. So, the product's own characteristics' direct effect on its markup can also falsify the model.

Second, consider the case where product characteristics are excluded from true marginal costs and the true conduct model has no direct effects of product characteristic's on markups, so that $\gamma_0 = 0$ and $\frac{\partial \Delta_{0t}}{\partial x_t} = 0$.[18] If the researcher does not know that characteristics are excluded from marginal costs, then only rival characteristics can be used as instruments. Here, falsification with rival product characteristics requires off-diag $\left\{\frac{\partial \Delta_{mt}}{\partial x_t}\right\} \neq 0$, so any model where a product's characteristics directly affect another product's markup will be falsified. If the researcher knows that $\gamma_0 = 0$, then both own and rival product characteristics can be used as instruments. Falsification then requires $\frac{\partial \Delta_{mt}}{\partial x_t} \neq 0$, so any model where at least one product's characteristic has a direct effect on at least one product's markup will be falsified.

Example 5: Return to our simple environment from Example 2, and consider a setting where the true model is Keystone and the researcher knows characteristics are excluded from marginal cost. Suppose the researcher's model is Cournot. Since both the Keystone and Cournot models have diagonal pass-through, rival cost shifters cannot falsify Cournot. The researcher does not know the effects of pure cost shifters on own costs, so own cost shifters cannot be used as instruments. On the other hand, own product characteristics can be used to falsify Cournot. To see this, note that we can simplify markups to $\Delta_{Cjt} = \frac{1}{\alpha}\left(1 + \exp(x_{jt}\beta - \alpha p_{jt})\right)$, which makes it clear that $\frac{\partial \Delta_{Cjt}}{\partial x_{jt}} = \frac{\beta}{\alpha}\exp(x_{jt}\beta - \alpha p_{jt})$, while for $\ell \neq j$, $\frac{\partial \Delta_{Cjt}}{\partial x_{\ell t}} = 0$. In short, the Cournot model is falsified because it implies that own characteristics have a nonzero causal effect on own prices (since $\beta \neq 0$), while the data show no causal effect.

Now suppose instead that the researcher's model is marginal cost pricing. Both Keystone and marginal cost pricing feature no direct effects of product characteristics on markups, so marginal cost pricing cannot be falsified by product characteristic instruments. •

[18] As one example, the rule-of-thumb class admits no direct effects on markups.

Linear index sufficiency: Thus far, we have seen that the direct effects of the product characteristics on markups play an important role in falsification, along with the pass-through matrix. Here, we explore a special case where these two objects satisfy a surprising equivalence: the characteristics and prices satisfy a *linear index sufficiency* condition.

Remark 3. For any model m (potentially including $m = 0$) such that Δ_{mt} depends on (p_t, x_t) only through the linear demand index $\beta x_t - \alpha p_t$, we have $\frac{\partial \Delta_{mt}}{\partial x_t} = -\frac{\beta}{\alpha}\frac{\partial \Delta_{mt}}{\partial p_t}$ and therefore $\frac{\partial \Delta_{mt}}{\partial x_t} = -\frac{\beta}{\alpha}\left(I - P_{mt}^{-1}\right)$.[19]

When demand itself depends on a linear index term – as in the logit and nested logit models – the index sufficiency condition in Remark 3 is satisfied whenever markups depend on prices and characteristics only through their impact on the demand function and its derivatives. For example, this is the case with Bertrand, Cournot, joint profit maximization, double-marginalization, and common ownership conduct models, although the rule-of-thumb class notably does *not* satisfy linear index sufficiency.

Now suppose both model m and the true model satisfy linear index sufficiency. When γ_0 is unknown, we can use rival product characteristics as instruments but not own product characteristics: $z_{jt} = x_{\ell t}$ for any $j \neq \ell$. The falsification condition in Proposition 1 can then be reduced to

$$E\left[e_j'\Big((\alpha\gamma_0 - \beta)P_{mt}^{-1}P_{0t} - (\alpha\gamma_m - \beta)I\Big)e_\ell \mid \mathrm{w}_{jt}, z_{jt}\right] \neq 0 \quad w.p.p.$$

for $j \neq \ell$.[20] Falsification then requires off-diag $\left\{(\alpha\gamma_0 - \beta)P_{mt}^{-1}P_{0t} - (\alpha\gamma_m - \beta)I\right\} \neq 0$, which arises if and only if $P_{mt}^{-1}P_{0t}$ is not diagonal. This is the same requirement as the rival cost shifter instruments.

We obtain a similar result when γ_0 is known. In that case, the researcher sets $\gamma_m = \gamma_0$ and can use both own and rival product characteristics as instruments. The falsification condition in Proposition 1 then reduces to

$$E\left[(\alpha\gamma_0 - \beta)e_j'\left(P_{mt}^{-1}P_{0t} - I\right)e_\ell \mid \mathrm{w}_{jt}, z_{jt}\right] \neq 0 \quad w.p.p.$$

for any (j, ℓ). So, falsification requires $P_{mt}^{-1}P_{0t} - I \neq 0$, which arises if and only if $P_{mt} \neq P_{0t}$. This is the same falsification requirement as the combination of own and rival cost-shifter instruments.

Remark 4. If product characteristics satisfy the linear index restriction in both model m and

[19] This last identity arises because $P_{mt} \equiv \left(I - \frac{\partial \Delta_{mt}}{\partial p_t}\right)^{-1}$ implies $\frac{\partial \Delta_{mt}}{\partial p_t} = I - P_{mt}^{-1}$.

[20] Here, we have multiplied through by α in the falsifiability condition to simplify the presentation.

the true model, then product characteristic instruments have the same falsifying content as cost shifter instruments.

Remark 4 provides a cautionary note: product characteristic instruments are not guaranteed to have different empirical content than cost shifter instruments. In order for these two types of instruments to be distinct, linear index sufficiency must be violated. For example, product characteristics that have their own random coefficients in the standard "BLP" demand system (Berry, Levinsohn, and Pakes, 1995) violate linear index sufficiency, giving them distinct empirical content.

Recap: Altogether, our analysis here suggests that product characteristics can be expected to have unique empirical content as instruments due to their direct effects on markups. However, there are special cases where their empirical content is equivalent to that of other instruments.

3.3 Tax Rate Instruments

Thus far, our analysis has focused on settings where cost shifters and/or product characteristics can be used as instruments. Now, we consider another type of instrument: tax rates. We assume that taxes vary at the market level, as is often the case in practice; they are not product-specific.[21] The government levies some combination of a unit tax τ_t and ad valorem tax v_t. Instead of working with v_t directly, we will work with ν_t, where $\frac{1}{\nu_t}$ is the fraction of the consumer's payment received by the firm. The form of ν_t depends on the remittance structure: it is equal to $1 + v_t$ when the tax is remitted by consumers and $1/(1 - v_t)$ when remitted by firms. Assuming taxes are fully salient and there is no avoidance or evasion, the after-tax version of Equation (1) is

$$\frac{1}{\nu_t} p_{jt} \quad = \quad \frac{1}{\nu_t} \Delta_{0jt} + \tau_t + c_{0jt}. \tag{4}$$

Rearranging into $\frac{1}{\nu_t}\left(p_{jt} - \Delta_{0jt}\right) = \tau_t + c_{0jt}$, we can interpret Equation (4) as setting after-tax marginal revenue equal to after-tax marginal cost.[22] For example, under the Keystone model markups are $\Delta_{Kjt} = \frac{1}{2} p_{jt}$. In the presence of an ad valorem tax, $\frac{1}{\nu_t} p_{jt} = 2\left(c_{Kjt} + \tau_t\right)$; the firm sets price such that after-tax marginal revenue is twice the after-tax marginal cost.[23]

We continue to maintain the rest of Assumption 1, so we assume that taxes do not

[21] Tax rates vary at the market level in our application.

[22] Without taxes, Equation (1) sets $p_{jt} - \Delta_{0jt} = c_{0jt}$; the right-hand side is marginal cost, so the left-hand side represents marginal revenue.

[23] We assume that firms treat the unit tax as an increase in cost and the ad valorem tax as a decrease in revenue.

directly enter the implied markup function Δ_m for any model we consider, including the true one. Additionally, taxes are excluded from the marginal cost function. Applying the Implicit Function Theorem to (4),[24] the causal effect of the unit tax on equilibrium prices is

$$\frac{\mathrm{d}p_{0t}}{\mathrm{d}\tau_t} = \nu_t P_{0t} \iota$$

where ι is a vector of ones. The causal effect of the ad valorem tax is

$$\frac{\mathrm{d}p_{0t}}{\mathrm{d}\nu_t} = \frac{1}{\nu_t} P_{0t} \left(p_t - \Delta_{0t}\right).$$

If variation in one or both tax rates is available, and assumed to be exogenous, then the tax rates can be used as instruments: $z_{jt} \subseteq \{\tau_t, \nu_t\}$. Re-writing ω_{mt} as $\omega_{mjt} = \frac{1}{\nu_t}\Delta_{0jt} - \frac{1}{\nu_t}\Delta_{mjt} - \tau_t + \bar{c}_{0j}(\mathrm{w}_{jt}) - \bar{c}_{mj}(\mathrm{w}_{jt}) + \omega_{0jt}$, falsification depends on the impossibility of finding a marginal cost function $\bar{c}_{mj}$ such that $E[\omega_{mjt}|\mathrm{w}_{jt}, z_{jt}] = 0$. Since $E[\omega_{0jt}|\mathrm{w}_t, z_{jt}] = 0$ and the tax instruments are excluded from the marginal cost functions $\bar{c}_{0j}(\cdot)$ and $\bar{c}_{mj}(\cdot)$, falsification occurs if $E[\frac{1}{\nu_t}(\Delta_{0jt} - \Delta_{mjt})|\mathrm{w}_{jt}, z_{jt}]$ varies with the tax rates. This leads to an analogue of Proposition 1 for tax instruments.

Proposition 2. *Under Assumptions 1-5, a model m is falsified by tax instruments $z_{jt} \subseteq \{\tau_t, \nu_t\}$ if and only if every admissible marginal cost function $\bar{c}_m(\cdot)$ is such that*

$$E\left[\frac{1}{\nu_t}\left(P_{mt}^{-1}\right)_j \left(\frac{\mathrm{d}p_{0t}}{\mathrm{d}z_{jt}} - \frac{\mathrm{d}p_{mt}}{\mathrm{d}z_{jt}}\right) \mid \mathrm{w}_{jt}, z_{jt}\right] \neq 0 \qquad w.p.p.$$

Proposition 2 establishes that failure to reproduce the true causal effect of the tax instruments on prices is necessary for falsification. Furthermore, taxes are excluded from marginal cost functions for all $j \in (1, \ldots, J)$, so the falsification condition is independent of $\bar{c}_m$ and reduces to $P_{mt}^{-1}\left(\frac{\mathrm{d}p_{0t}}{\mathrm{d}z_t} - \frac{\mathrm{d}p_{mt}}{\mathrm{d}z_t}\right) \neq 0$. Because P_{mt}^{-1} is full rank, this condition holds if and only if

$$\frac{\mathrm{d}p_{mt}}{\mathrm{d}z_t} \neq \frac{\mathrm{d}p_{0t}}{\mathrm{d}z_t}.$$

So, any conduct model that fails to replicate the true causal effect of tax variation on prices will be falsified. To see what this requires for each tax, we substitute the causal effect formulas derived above. For the unit tax, $\frac{\mathrm{d}p_{0t}}{\mathrm{d}\tau_t} = \nu_t P_{0t}\iota$ and $\frac{\mathrm{d}p_{mt}}{\mathrm{d}\tau_t} = \nu_t P_{mt}\iota$, so falsification requires

[24] Define the vector $\tilde{c}_t(\tau, \nu) \equiv \nu c_{0t} + \nu\tau\iota$. Then, we can re-write (4) in vector form as $p - \Delta_0(p, x_t, \xi_t) - \tilde{c}_t(\tau, \nu) = 0$. $p_{0t}(\tau, \nu)$ is implicitly defined as the solution to this equation. For tax rate z, the Implicit Function Theorem implies $\frac{\mathrm{d}p_{0t}}{\mathrm{d}z_t} = P_{0t}\frac{\partial \tilde{c}_t}{\partial z_t}$. For the unit tax, $\frac{\partial \tilde{c}_t}{\partial \tau_t} = \nu_t\iota$. For the ad valorem tax, $\frac{\partial \tilde{c}_t}{\partial \nu_t} = (c_{0t} + \tau_t\iota)$, where we have $c_{0t} + \tau_t\iota = \frac{1}{\nu_t}(p_t - \Delta_{0t})$ from (4) in vector form.

$P_{mt}\iota \neq P_{0t}\iota$; that is, the row sums of the two pass-through matrices must differ. For the ad valorem tax, $\frac{\mathrm{d}p_{0t}}{\mathrm{d}\nu_t} = \frac{1}{\nu_t}P_{0t}(p_t - \Delta_{0t})$ and $\frac{\mathrm{d}p_{mt}}{\mathrm{d}\nu_t} = \frac{1}{\nu_t}P_{mt}(p_t - \Delta_{mt})$, so falsification requires $P_{mt}(p_t - \Delta_{mt}) \neq P_{0t}(p_t - \Delta_{0t})$, which depends on both the pass-through matrix and the level of markups.

Remark 5. The unit tax rate instrument targets row sums of the pass-through matrix; falsification requires $P_{mt}\iota \neq P_{0t}\iota$. The ad valorem tax rate instrument targets the product of the pass-through matrix and the vector of differences between prices and markups; falsification requires $P_{mt}\left(p_t - \Delta_{mt}\right) \neq P_{0t}\left(p_t - \Delta_{0t}\right)$.

The unit tax rate instrument's empirical content is similar to the combination of own and rival cost shifter instruments, but it targets the row sums of the pass-through matrix instead of the full matrix. This is because a change in the unit tax shifts all marginal costs by an equal amount, while own and rival cost shifters produce separate variation in marginal costs. Consequently, the unit tax rate instrument can falsify important models of conduct, but there are some it cannot falsify.

Example 6: Return to the setting in Example 2 and suppose that we observe variation in a unit tax which is levied on both firms, τ_t, so that $z_t = \tau_t$. Let the true model be Keystone, with $P_{Kt} = 2I$, and let the model to be falsified be marginal cost pricing so that, with $P_{MCt} = I$. Falsification with the unit tax rate instrument occurs because the rows of P_{Kt} sum to 2 while the rows of P_{MCt} sum to 1. Now, suppose instead the true model implied a pass-through matrix of $P_{0t} = \left[\begin{smallmatrix} 0.8 & 0.2 \\ 0.2 & 0.8 \end{smallmatrix}\right]$. The rows of this matrix also sum to 1, so marginal cost pricing would not be falsified by the unit tax rate instrument in this case. •

While the unit tax rate instrument targets row sums of the pass-through matrix, the ad valorem tax rate instrument targets the product of the pass-through matrix with the vector of differences between prices and markups. This gives the ad valorem tax rate unique empirical content relative to the other instruments we study. To see why, suppose that the pass-through matrix for model m is identical to the truth: $P_{mt} = P_{0t}$. In that case, we have $P_{mt}\left(p_t - \Delta_{mt}\right) \neq P_{0t}\left(p_t - \Delta_{0t}\right)$ if and only if

$$\Delta_{mt} \neq \Delta_{0t}.$$

Thus, the ad valorem tax can still falsify model m when the levels of its implied markups differ from the true markups. We illustrate this in the next example.

Example 7: Return to the example environment, and suppose the true model is one in which firms set pre-tax markups to a constant level across markets (i.e., a fixed dollar amount above marginal cost), so that $\Delta_{0jt} = \zeta_{0j}$. We consider falsifying a model in which firms set constant

markups at a different level: $\Delta_{mt} = \zeta_m \neq \zeta_0$. Here, $P_{0t} = P_{mt} = I$ and instruments have no direct effect on markups. Product characteristics, rival cost shifters, own cost shifters, demand shifters, and unit tax rate instruments are all therefore irrelevant for falsification. However, the ad valorem tax rate instrument falsifies the model because $\Delta_{mjt} \neq \Delta_{0jt}$. Intuitively, although both models share the same cost pass-through ($P_{0t} = P_{mt} = I$), they imply different levels of after-tax revenue at any given tax rate. The ad valorem tax, which scales revenue multiplicatively, amplifies this difference in markup levels into different causal effects on prices. •

Example 7 presents a case where the ad valorem tax instrument falsifies a model that cannot be falsified by other conduct instruments. However, there are also settings where the ad valorem tax instrument cannot falsify a model that is falsified by other instruments. Consider, for example, the rule-of-thumb class where $\Delta_{\phi t} = \frac{\phi-1}{\phi} p_t$. In that case, $P_{\phi t} = \phi I$ and $p_t - \Delta_{\phi t} = \frac{1}{\phi} p_t$. So, we have $P_{\phi t}\left(p_t - \Delta_{\phi t}\right) = p_t$ which does not vary with ϕ. Consequently, the ad valorem tax instrument cannot falsify an incorrect rule-of-thumb model when the truth is another rule-of-thumb. In contrast, falsification can be obtained with the unit tax instrument because the row sums of the pass-through matrix are equal to ϕ; falsification then obtains when $\phi_m \neq \phi_0$.

Recap: The discussion above illustrates the usefulness of tax rates as instruments. Falsification depends on the conduct model's (in-)ability to replicate the tax rates' true causal effects on equilibrium prices. The pass-through matrix still plays a central role but is leveraged in different ways by the tax instruments, since one enters the equilibrium condition additively and the other enters multiplicatively. This results in substantially different empirical content for the tax rate instruments.

3.4 Counterfactual-Relevant Instruments

Researchers often use the model of conduct as a key input into counterfactual analyses. Different counterfactual exercises depend on different features of the conduct model, and our falsification framework can direct researchers toward instruments that target these particular features, making the counterfactual analysis more credible.

As an illustration, we consider the problem of learning the Laffer curve, which has garnered recent interest in applied work (Miravete et al., 2018; Hollenbeck and Uetake, 2021; Brugués and De Simone, 2026) and is also the subject of our empirical application in Section 5.[25] Counterfactual predictions are essential to understanding the Laffer curve, as taxes

[25] The insights in this section readily extend to other optimal tax policy settings: for example, if the government wished to maximize a convex combination of revenue and consumer surplus.

usually have limited support in the data. The policymaker must use the model to forecast revenue at unobserved tax rates. We limit attention to the optimal tax in a single market, t, for expositional parsimony. Government revenues in market t for any tax rates (ν, τ) are[26]

$$R_t(\nu, \tau) = \tau \sum_j s_{jt}\big(p_{0t}(\nu, \tau)\big) + \frac{\nu - 1}{\nu} \sum_j p_{0jt}(\nu, \tau) s_{jt}\big(p_{0t}(\nu, \tau)\big). \tag{5}$$

This expression emphasizes that market shares depend on prices, and prices depend on tax rates; all other primitives are held constant and suppressed. The observed price $p_t = p_{0t}(\nu_t, \tau_t)$ corresponds to the observed tax rates (ν_t, τ_t). The policymaker knows the revenue at observed tax rates, but must use a model m to predict counterfactual prices $p_{mt}(\nu, \tau)$ and therefore revenue at other rates. Recent literature (Miravete et al., 2018; Hollenbeck and Uetake, 2021) shows that the Laffer curve can differ substantially across conduct models. The policymaker, not observing the true conduct model, wants the model that best matches the true Laffer curve. Our framework sheds light on the link between falsification by particular instruments and the counterfactual Laffer curve.

We assume the government optimizes one tax rate, keeping the other fixed. Our application to the Washington marijuana market exemplifies this: the government only uses an ad valorem tax. We analyze the Laffer curves for unit and ad valorem tax rates separately.

Remark 6. For tax rate z, if $\frac{\mathrm{d}p_{mt}}{\mathrm{d}z} = \frac{\mathrm{d}p_{0t}}{\mathrm{d}z}$ for all z, then model m produces the true Laffer curve. For the unit tax, this requires $P_{mt}\iota = P_{0t}\iota$. For the ad valorem tax, it requires $P_{mt}\left(p_t - \Delta_{mt}\right) = P_{0t}\left(p_t - \Delta_{0t}\right)$.

The mechanics driving Remark 6 can be summarized as follows. All models are constructed to rationalize observed prices, so $p_{mt}(z_t) = p_{0t}(z_t) = p_t$. So, matching the derivatives implies that $p_{mt}(\cdot) = p_{0t}(\cdot)$ for all z. This means model m will produce the true Laffer curve, since the true pricing function $p_{0t}(\cdot)$ is the only unknown object in (5). The conditions specific to the unit and ad valorem taxes come from the formulas for their causal effects, provided in Section 3.3.

Together, Remarks 5 and 6 imply that the tax rate itself is the most relevant conduct instrument when the researcher is ultimately interested in the Laffer curve: it distinguishes conduct models through their implied causal effects on prices, and these causal effects are precisely what determine the counterfactual Laffer curve.

Remarks 5 and 6 also highlight the key features of the model's pass-through matrix and implied markups that drive the tax rate's causal effect on prices. Thus, researchers can di-

[26] By definition, $\frac{1}{\nu}$ is the firms' remaining share of revenue after the government takes its cut. So, the government's share is $1 - \frac{1}{\nu} = \frac{\nu - 1}{\nu}$.

rectly target the most relevant part of the pass-through matrix for the policy counterfactual. Our next two examples further illustrate this connection.

Example 8: Consider the simple environment of Example 2. Suppose the true model is Keystone pricing, and the alternative model is Bertrand. Additionally, suppose there is no ad valorem tax, and the researcher is interested in the Laffer curve for the unit tax. Existing variation in the unit tax falsifies Bertrand because $P_{Bt}\iota < 1 < 2 = P_{Kt}\iota$. Further, since $P_{Bt}\iota \neq P_{Kt}\iota$, the Laffer curve under Bertrand differs from the truth. Now consider falsifying a shifted Keystone model ($m = S$) with $p_{jt} = 2c_{jt} + \zeta_{jt}$, where ζ_{jt} is a constant. Notice that, since $P_{St} = P_{Kt}$ and therefore $P_{St}\iota = P_{Kt}\iota$, the shifted Keystone model is not falsified by the unit tax instrument. However, the researcher can still learn the true Laffer curve with this model because the pass-through matrices are the same. •

Example 9: Now suppose there is no unit tax and the researcher wants the Laffer curve for the ad valorem tax. The true model is still Keystone. Keystone pricing is a special case of the more general rule-of-thumb class where $P_{\phi t}\left(p_t - \Delta_{\phi t}\right) = p_t$ for any ϕ, as shown in Section 3.3.[27] No other model in the rule-of-thumb class can be falsified by the ad valorem tax instrument, since $P_{\phi t}\left(p_t - \Delta_{\phi t}\right) = p_t$ does not vary across rule-of-thumb models. But this is not an issue for the researcher's purposes, since all rule-of-thumb models will also produce the correct Laffer curve. •

Sometimes, the data does not contain any variation in the tax rate of interest. In that case, a researcher can use our framework to understand the tradeoffs involved in selecting another instrument (i.e., the other tax rate, rival cost shifters, or product characteristics) to learn conduct and, subsequently, the Laffer curve. These instruments target other features of the pass-through matrix that may not be first-order relevant to the tax counterfactual. They may fail to falsify conduct models that deliver incorrect Laffer curves; or conversely, they may falsify models that deliver the correct Laffer curve.

For example, suppose that the researcher wants the Laffer curve for the unit tax, and that there is a cost shifter w_t such that $\bar{c}_0\left(\mathrm{w}_{jt}\right) = \gamma^{\text{const}} + \gamma_0 \mathrm{w}_{jt}$ where the coefficient γ_0 is known by the researcher *a priori*. This is the case where both own and rival cost shifters are valid instruments, which target the full pass-through matrix (Remark 2). So, these instruments falsify any model with $P_{mt} \neq P_{0t}$ even though we only require $P_{mt}\iota = P_{0t}\iota$ to obtain the true unit tax Laffer curve. Thus, the own and rival cost instruments are "second best" in the sense that they are overkill; they falsify models that produce incorrect Laffer curves, but they can also falsify models that produce the correct Laffer curve.

Interestingly, it is possible to emulate the first-best instrument (the unit tax itself) in

[27] As noted in Example 1, Keystone and marginal cost pricing correspond to $\phi = 2$ and $\phi = 1$, respectively.

this setting. To that end, let $\bar{\mathrm{w}}_t \equiv \frac{1}{J}\sum_{j=1}^{J} \mathrm{w}_{jt}$ be the average value of the cost shifter and define $\tilde{\mathrm{w}}_{jt} \equiv \mathrm{w}_{jt} - \bar{\mathrm{w}}_t$. Then, we can write $\bar{c}_0\left(\mathrm{w}_{jt}\right) = \gamma^{\text{const}} + \gamma_0 \bar{\mathrm{w}}_t + \gamma_0 \tilde{\mathrm{w}}_{jt}$, and we can see that changing $\bar{\mathrm{w}}_t$ shifts all products' marginal costs by γ_0 simultaneously. Using $\bar{\mathrm{w}}_t$ as the instrument, the falsification condition in Proposition 1 then reduces to

$$E\left[\gamma_0 \left(P_{mt}^{-1}\right)_j \left(P_{0t}\iota - P_{mt}\iota\right) \mid \tilde{\mathrm{w}}_t, \bar{\mathrm{w}}_t\right] \neq 0$$

because γ_0 is already known by the researcher, who imposes $\gamma_m = \gamma_0$. This is valid for each $j \in \{1, \ldots, J\}$, so falsification with $\bar{\mathrm{w}}_t$ as the instrument requires $P_{mt}\iota \neq P_{0t}\iota$; only models that produces incorrect Laffer curves are falsified.

The situation changes when the researcher is focused on the Laffer curve for an ad valorem tax instead of a unit tax. Both the pass-through and the level of markups determine the ad valorem Laffer curve, and no instrument other than ad valorem taxes targets both of these objects. Which instrument is second-best in this case depends on the relative importance of indirect and level effects, which will vary by application.

Recap: Researchers can use our falsification framework to understand the tradeoffs inherent to various instruments when their ultimate goal is to perform counterfactual analysis. In the case of Laffer curves, the tax rate itself is the first-best instrument. When historical variation in the tax rate is unavailable, researchers may be able to use our framework to replicate the first-best instrument; we demonstrated this for the unit tax using average cost shifters. Replicating the ad valorem tax instrument is more challenging, as it requires targeting both pass-through and markup levels simultaneously. The availability of such replica instruments depends on the counterfactual at hand.

3.5 Summary of IV Relevance in Examples

To help the reader, we now summarize the takeaways from the examples in the paper. Recall that we consider a stylized environment with two single-product firms, logit demand, and no unobservable variation in demand and cost. In the following table, for select combinations of true and alternative model, we indicate which of the sources of variation considered in the examples will permit falsification (✓), and which ones will not (×).

4 Implications for Empirical Practice

Our analysis in Sections 2 and 3 has important practical implications for researchers performing inference on conduct. In this section, we link instrument relevance to finite-sample

Table 1: Summary of IV Relevance in Examples

		Cost Shifters		Product Characteristics		Taxes	
		Rival	Own+Rival	Rival	Own+Rival	Unit	Ad Valorem
True	Tested	$\frac{(P_{mt})_{j\ell}}{(P_{mt})_{\ell\ell}} \neq \frac{(P_{0t})_{j\ell}}{(P_{0t})_{\ell\ell}}$	$P_{mt} \neq P_{0t}$	off-diag $P_{mt}^{-1} P_{0t} \frac{\partial \Delta_0}{\partial x} - \frac{\partial \Delta_m}{\partial x}$	$P_{mt}^{-1} P_{0t} \frac{\partial \Delta_0}{\partial x} \neq \frac{\partial \Delta_m}{\partial x}$	$P_{mt\iota} \neq P_{0t\iota}$	$P_{mt}(p-\Delta_{mt}) \neq P_{0t}(p-\Delta_{0t})$
Bertrand	Keystone	✓	✓	✓	✓	✓	✓
	Cournot	✓	✓	✓	✓	✓	✓
	MC	✓	✓	✓	✓	✓	✓
Cournot	Keystone	×	✓	×	✓	✓	✓
	MC	×	✓	×	✓	✓	✓
Keystone	MC	×	✓	×	×	✓	×
$\Delta_0 = \zeta_0$	$\Delta_m = \zeta_m$	×	×	×	×	×	✓

The table summarizes falsification results from the examples in the paper. The environment is two single-product firms with logit demand. Each cell indicates whether the instrument can (✓) or cannot (×) falsify the tested model when the true model generates the data. The row below the instrument labels shows the key falsification condition from Remarks 1–5, simplified by setting $\gamma_0 = 0$ for product characteristics. The "Rival" columns assume γ_0 is unknown; the "Own+Rival" columns assume γ_0 is known.

inference, focusing on statistical degeneracy concerns (Backus et al., 2021; Duarte et al., 2024). We then discuss the implications of our results for demand estimation, arguing that the flexibility of demand *curvature* in the researcher's model is crucial for accurate inference on conduct. Finally, we summarize supporting Monte Carlo evidence.

Our main econometric results are derived at the population level, but empirical practice requires finite sample inference. There are two main approaches: testing and estimation. Testing performs pairwise model comparisons between candidate models that may be non-nested (e.g., Duarte et al., 2026), while estimation considers a continuum of models within a nested class and seeks to point-identify the best fit (e.g., Miller and Weinberg, 2017). We focus on the testing approach in our discussion here, since it is a natural fit for the non-nested model comparisons from our empirical application in Section 5. The main insights of our discussion readily extend to estimation, which we discuss in more detail in Appendix C.

We employ the model selection test of Rivers and Vuong (2002) (RV), which has become the standard approach for conduct testing in differentiated products markets (Backus et al., 2021; Duarte et al., 2024). The RV test's null hypothesis states that two competing models of conduct $m = 1, 2$ have the same lack-of-fit: $H_0^{\text{RV}} : Q_1 = Q_2$, while the alternatives correspond to superior fit of one of the two models: $H_1^{\text{RV}} : Q_1 < Q_2$ and $H_2^{\text{RV}} : Q_2 < Q_1$. With this formulation, the testing procedure determines which of the two models has the smallest lack of fit.

As in Duarte et al. (2024), we express Q_m as a GMM objective function $Q_m = g_m' W g_m$,

where $g_m = E[z_{jt}\omega_{mjt}]$ and $W = E[z_{jt}z'_{jt}]^{-1}$ is the weight matrix. For this GMM measure of fit, the RV test statistic is then

$$T^{\text{RV}} = \frac{\sqrt{n}(\hat{Q}_1 - \hat{Q}_2)}{\hat{\sigma}_{\text{RV}}},$$

where $\hat{Q}_m = \hat{g}'_m \hat{W} \hat{g}_m$, $\hat{g}_m = n^{-1}\hat{z}'\hat{\omega}_m$, $\hat{W} = n(\hat{z}'\hat{z})^{-1}$, and $\hat{\sigma}^2_{\text{RV}}$ is an estimator for the asymptotic variance of $\sqrt{n}(\hat{Q}_1 - \hat{Q}_2)$.[28]

4.1 Instrument Relevance and (Non-)Degeneracy of the Test

There is a tight connection between our falsification framework and the RV test, which can suffer from statistical degeneracy problems identified in Duarte et al. (2024). Crucially, our theoretical results enable researchers to select instruments that will avoid degeneracy, thereby ensuring reliable inference. Consider a researcher with a menu of candidate models. For any pair of models in this menu, the RV test statistic will be degenerate – leading to unreliable inference – if and only if both models are not falsified by the chosen instruments. Our theoretical results allow researchers to avoid this problem by selecting instruments that will falsify at least one model in each pair being tested.

Specifically, suppose the researcher wants to test models 1 and 2, and, for simplicity, further suppose that model 1 is the true model of conduct. Our results in Section 3 show that avoiding degeneracy requires finding instruments z_{jt} such that the pass-through matrices P_{1t} and P_{2t} differ in ways that the instruments can detect. For instance, if both models have diagonal pass-through matrices - or more generally, pass-through matrices with identical ratios of off-diagonal to diagonal elements - then rival cost shifters will be irrelevant for falsifying model 2. In this case, $Q_1 = Q_2 = 0$ and the RV test statistic is degenerate. However, product characteristics or tax instruments may be relevant to falsify model 2; forming the RV test statistic with these sources of variation would therefore avoid degeneracy. By checking these conditions ex-ante, researchers can ensure they select instruments that will produce informative test statistics.

Moreover, when multiple models are under consideration, our framework guides the construction of model confidence sets. If the researcher has models $\{1, 2, ..., M\}$ and wishes to determine which are consistent with the data, they can use our results to select instruments that are relevant for all pairwise comparisons. The F-statistic diagnostic from Duarte et al. (2024) then provides finite-sample evidence of whether the selected instruments achieve sufficient relevance.

[28] See Duarte et al. (2024) for a variance estimator accounting for randomness in $\hat{W}$ and $\hat{g}_m$.

Our analysis in Section 3 also suggests that degeneracy concerns can be alleviated when the researcher has a particular counterfactual analysis in mind. In that case, researchers can use our framework to determine the first-best, counterfactual-relevant instrument as in Section 3.4. When the researcher uses this instrument, the test will only be degenerate when comparing conduct models that produce the same counterfactual result. This instrument makes the test robust to degeneracy, in the sense that it only becomes unreliable when the differences between the conduct models are irrelevant for the researcher's purposes. We therefore recommend that researchers use the first-best, counterfactual-relevant instrument when it is available. We demonstrate this practice in our empirical application in Section 5, where our objective is to trace out the Laffer curve for the ad valorem tax.

Recap: Our econometric framework can guide instrument selection in the RV test, so that researchers can perform tests that are robust to degeneracy issues or avoid them altogether.

4.2 Demand Estimation and Pass-Through

A key theme arose in Section 3: the cost pass-through matrix plays a central role in falsification. Cost pass-through for many conduct models, in turn, depends crucially on the curvature of demand (Bulow and Pfleiderer, 1983; Weyl and Fabinger, 2013; Miravete et al., 2026).

This has substantial implications for demand estimation. Thus far, we have taken the researcher's estimated demand model as given and have assumed it reflects the true demand function. In practice, researchers often use demand systems that rely on simplifying assumptions and parsimonious parametric restrictions, using a combination of economic intuition and "sanity checks" to inform accuracy. In much of the literature on demand estimation, the focus is mainly on accurately capturing substitution patterns, i.e., average elasticities. Our analysis suggests that this is not enough for accurate conduct inference: the researcher must also accurately capture demand *curvature*.

The parsimonious demand models used in empirical practice can implicitly place substantial restrictions on curvature, such as log-concavity (Miravete et al., 2026). Restrictions on curvature translate into restrictions on cost pass-through, potentially biasing inference on firm conduct.

As a concrete example, consider a researcher who tests for collusion among firms in the form of joint profit maximization against (non-collusive) Cournot competition. Similar to recent literature using tax rate instruments to test for collusion (Brugués and De Simone, 2026), the researcher uses a unit tax as the conduct instrument. Furthermore, suppose that the true demand is log-convex and firms do, in fact, engage in joint profit maximization. This results in "over-shifting" of the tax where prices rise by more than one dollar for each

dollar of unit tax imposed (Weyl and Fabinger, 2013; Miravete, Seim, and Thurk, 2023), so that $(P_{0t})_j \iota > 1$ for each j.

Now suppose the researcher uses a log-concave demand model, such as the popular logit or nested logit (Miravete et al., 2026). In the two-good logit setting from Section 3, the pass-through of the unit tax into prices satisfies

$$P_{JPt}\iota = \begin{bmatrix} s_{0t} \\ s_{0t} \end{bmatrix} \quad < \quad P_{Ct}\iota = \begin{bmatrix} \frac{s_{0t}}{1-s_{2t}} \\ \frac{s_{0t}}{1-s_{1t}} \end{bmatrix} \quad < \quad \begin{bmatrix} 1 \\ 1 \end{bmatrix},$$

where $s_{0t} = 1 - s_{1t} - s_{2t}$. Here, both models predict under-shifting (Miravete et al., 2023), with greater market power implying lower pass-through (Weyl and Fabinger, 2013). Since the true DGP exhibits over-shifting ($P_{0t}\iota > \iota$), we then have $P_{JPt}\iota < P_{Ct}\iota < P_{0t}\iota$.

In Section 3, we showed that the unit tax's empirical content as a conduct instrument comes from its causal effects on prices: models with $P_{mt}\iota = P_{0t}\iota$ will not be falsified. Intuitively, models with pass-through closer to the truth will likely be favored in finite-sample conduct tests. So, the researcher's conduct test favors Cournot even though the true conduct is collusive joint profit maximization.

The issue of limited demand curvature extends well beyond the logit and nested logit. Birchall et al. (2024) and Miravete et al. (2026) show that the ubiquitous "BLP" random coefficients logit model (Berry et al., 1995) can also impose important curvature limitations under common distributional assumptions on consumer taste heterogeneity, implicitly limiting model-implied cost pass-through. Miravete et al. (2026) propose alternative distributional forms that facilitate more flexibility in cost pass-through.

In practice, we recommend that researchers estimate demand forms that admit a wide variety of curvatures when performing inference on conduct, such as those in Miravete et al. (2026). Increased flexibility of the demand model, however, places a higher burden on the data in estimation. When the variation in demand instruments is too limited to fully identify the flexible models of Miravete et al. (2026), we recommend that researchers perform robustness checks by testing conduct with a variety of demand parameter values. We demonstrate this approach in our empirical application in Section 5, where we find that the results of our conduct tests are robust to a wide variety of demand curvatures.

Recap: Pass-through plays a central role in conduct inference, and demand curvature is a key determinant of pass-through for many conduct models. Insufficient flexibility in demand curvature can bias conduct inference. Researchers should estimate demand models with flexible curvature when the data contains sufficient identifying variation, or perform robustness checks across curvature specifications when such variation is unavailable in the data.

4.3 Summary of Monte Carlo Simulations

To demonstrate the practical applicability of our theoretical insights, we conduct Monte Carlo simulations that illustrate how different instruments perform in testing various conduct models. Full details of the simulation design and results appear in Appendix D. We provide a brief summary here.

We simulate data for markets with characteristics closely matching those in our empirical application, with the true conduct being Keystone pricing. Our simulations confirm the key insights from Section 3. When testing Bertrand against Keystone, all instruments – unit taxes, ad valorem taxes, rival cost shifters, and product characteristics – successfully reject the false model, consistent with our theoretical predictions. However, when testing Cournot against Keystone, rival cost shifters fail to provide power, as predicted by our analysis showing that both models have diagonal pass-through matrices under logit demand. Importantly, the simulations also demonstrate that for tax policy counterfactuals, using the appropriate tax instrument leads to correct inference about the optimal tax rate. Other instruments may falsify models that nonetheless yield the correct Laffer curve, or fail to falsify models that generate incorrect Laffer curves.

5 Application: Conduct and the Laffer Curve in the Washington Recreational marijuana Industry

As an application of our framework, we consider the problem of setting the optimal ad valorem tax in Washington State's marijuana market, which is well-suited for several reasons. First, setting the optimal tax is an important consideration in this market, as revenues are substantial. Second, optimal taxes depend on conduct (Miravete et al., 2018; Hollenbeck and Uetake, 2021), and it is unclear how well retailers' conduct in this market aligns with standard structural models (Escudero, 2018; Hollenbeck et al., 2026). Third, our results suggest that for a tax exercise, conduct would best be learned using an ad valorem tax instrument, which is available in this market through local marijuana tax variation.

5.1 Institutional Background and Data

Washington legalized marijuana for medical use in 1998, and in 2012, through voter approval and passage of I-502, was among the first states to legalize recreational marijuana.[29] In July of 2014, retail sales of marijuana began in Washington.

[29] As of 2024, 38 states have legalized marijuana for medical use and 24 for recreational use.

The state regulates the marijuana industry in several ways. First, there is a cap on the number of retailers.[30] Second, the state restricts vertical relations between producers (growers), processors, and retailers. While producers and processors — those who convert marijuana plants into various products — can integrate with one another, neither can integrate with a retailer. Only retailers can sell marijuana products directly to consumers. Third, the state assesses an ad valorem tax on marijuana sales. When marijuana was first legalized, the state levied a 25% tax at each point of sale — from producers to processors, processors to retailers, and retailers to consumers. In July 2015, this changed to a 37% tax on the final sale to consumers only. Additionally, local municipalities imposed their own ad valorem taxes, which range from seven to ten percent.

In part to enforce tax collection, the state tracks detailed "seed-to-sale" data on all marijuana products. Our empirical application benefits from these unusually detailed data. We construct a dataset containing product shares and prices for many city-month markets. The underlying data come from BioTrack, an administrative data collection software that tracks marijuana from plant production and processing through retail ("seed-to-sale"). These data include the full universe of wholesaler and retailer transactions for marijuana products. Thus, for every transaction, we observe prices paid both by consumers to retailers as well as by retailers to wholesalers. Biotrack also collects supplemental information on the organizations in the market, from which we observe when retailers began operating, and lab testing of plants, from which we observe marijuana product CBD and THC content.

Due to the difficulty of comparing package size weights between different product types (e.g., liquid, extracts, and solids), we focus on products labeled as "usable" marijuana (dried leaves and flowers), which represent 72% of all transactions in the data. Within this category, we subset to package sizes of 1 and 3.5 grams (following Escudero, 2018), which are by far the most popular, accounting for 60% of usable product revenue.[31] Features of the marijuana market complicate the observed prices in the raw data. Because marijuana is illegal at the federal level, retailers accept cash payments and manually record transactions, resulting in inconsistencies regarding whether reported prices include the tax. Hollenbeck and Uetake (2021) correct for this, and we use their reconstructed tax-inclusive retail prices.[32]

We define a product to be a package size from a processor sold at a retailer in a month, and markets to be city-month pairs. We calculate a product's price as the sales volume-weighted

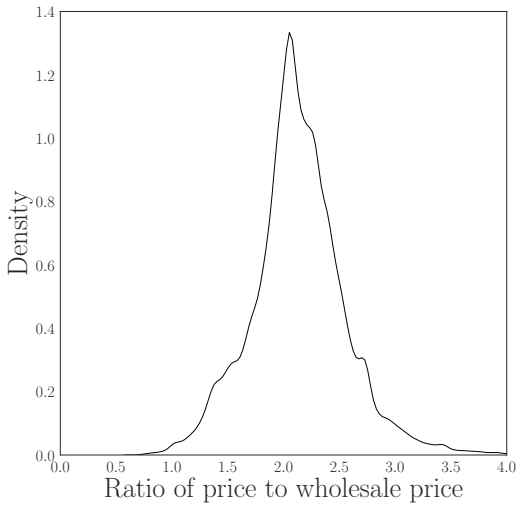
[30] The Washington State Liquor and Marijuana Board initially limited the number of retailer licenses to 334, and allocated licenses across counties proportional to population. In jurisdictions with too many applications, licenses were assigned via lottery. In January 2016, the state increased the number of retail licenses to 556. The state also caps the number of each licensee's retail locations at three.

[31] See Table 5 in Appendix E for a summary of all data cleaning steps.

[32] Additional details on data construction can be found in Appendix A of Hollenbeck and Uetake (2021).

average price. We define the market size to be twice the maximum amount ever purchased in a given geographic market. We further draw 2000 consumers' incomes from the empirical distribution of income in each city, which we obtain from the US Census. The data include prices and shares for 1,428 products across 2,639 markets from August 2014 to May 2017.

FIGURE 1: Ratio of Retail and Wholesale Price



The figure plots densities of the ratio of retail and wholesale prices for each product in our data.

Appendix Table 6 displays summary statistics of the data used in estimation. Wholesalers charge an average of \$3.60 for a gram of marijuana, while retailers charge \$7.60, on average. While the average market has over 100 marijuana processors, retail market quotas restrict the average market to 14 retailers. We follow the literature (Escudero, 2018; Hollenbeck and Uetake, 2021) in concluding from these facts that wholesaler market power is unlikely to be important, and we focus our attention on how retailers set prices instead.

Figure 1 graphs the ratio of retail unit prices to wholesale unit prices. We see a mass of ratios concentrated around two, lending credence to the hypothesis that firms engage in Keystone pricing. A formal test follows in Section 5.3.

5.2 Demand Estimation

We follow Hollenbeck and Uetake (2021) in adopting a random coefficient nested logit demand model (Grigolon and Verboven, 2014). We briefly summarize our demand model here, and provide a more detailed exposition in Appendix E.

Product characteristics include a constant, package size, THC and CBD levels (and their values squared), and the log of the number of products offered in the store to capture variation in shelf space across stores. We allow consumer preferences for characteristics and prices to

vary with individual-level income. We also include fixed effects at the retailer, processor, and year-month levels. Our nesting structure includes all inside goods in one nest.

Following Hollenbeck and Uetake (2021) we construct demand instruments from several sources of exogenous variation. These include the number of products sold at competing dispensaries in the market, also interacted with the mean income, BLP-style instruments (sum of THC and CBD for products within a store, and for products in all other stores), and exogenous own cost shifters (rainfall and temperature in the region of production and their lags). We estimate demand using `PyBLP` (Conlon and Gortmaker, 2020).

Results for demand estimation are reported in Appendix Table 7. Similar to Hollenbeck and Uetake (2021), we find a positive interaction between income and price and a negative interaction between income and the constant. We also find that consumers prefer products with a higher level of THC, though at a declining rate, while they dislike products with higher levels of CBD. The median own-price elasticity in our sample is -6.45, suggesting a fair degree of substitution across strains of usable marijuana.

While these product-level elasticities are high in magnitude, this reflects our finer product definition relative to Hollenbeck and Uetake (2021), who aggregate all strains within a category (usable, liquid edible, solid edible, and extract): we define a product as a package size from a processor sold at a retailer in a month, focusing exclusively on usable marijuana. At the aggregate level, where substitution is measured across a comparable degree of product aggregation, our elasticity of -3.32 (-2.90 when weighted by market size) is in line with Hollenbeck and Uetake (2021) (-2.89) and broadly similar to estimates for the Pennsylvania liquor market in Miravete, Seim, and Thurk (2020) (-2.48). Because the aggregate elasticity is a key input for our conduct tests and counterfactuals, in Appendix F we further verify robustness by calibrating demand to alternative aggregate elasticity levels and find that our testing results are unaffected.

One potential concern is that our implementation of the RCNL model restricts the range of curvature of demand (Miravete et al., 2026). In turn, misspecifying demand in this way can lead to restrictions on the pass-through implied by certain models of conduct (e.g., Bertrand). Because we show, in the previous section, that differences in pass-through matrices help distinguish different models of conduct, this potential misspecification could affect our test results. Miravete et al. (2026) allow for greater flexibility in modeling heterogeneous price sensitivity by incorporating a Box-Cox transformation of the income interaction. They show that this can greatly increase the range of curvature implied by discrete choice demand models. Without microdata, we lack sufficient identifying power to pin down the additional parameters required by this approach. Instead, in Appendix F, we calibrate different approximations for the Box-Cox transformations for income and include them in the RCNL

demand system. We find that our testing results are unaffected.

5.3 Test for Conduct

Models of Conduct: We consider three models of conduct that are discussed in previous literature for this market: (i) *Marginal Cost Pricing*—retailers choose tax-exclusive prices equal to marginal cost; (ii) *Keystone Pricing*—retailers choose tax-exclusive prices equal to twice marginal cost; and (iii) *Bertrand Pricing (Nash Price Setting)*—retailers set prices competitively to maximize profits in the complete information pricing game (considered in Hollenbeck and Uetake, 2021). Escudero (2018) provides some descriptive evidence that Keystone pricing is plausible in the market during our sample period, while Hollenbeck et al. (2026) provide evidence of strategic interactions in a later period. It is thus unclear *a priori* which model best fits the data during our sample period.

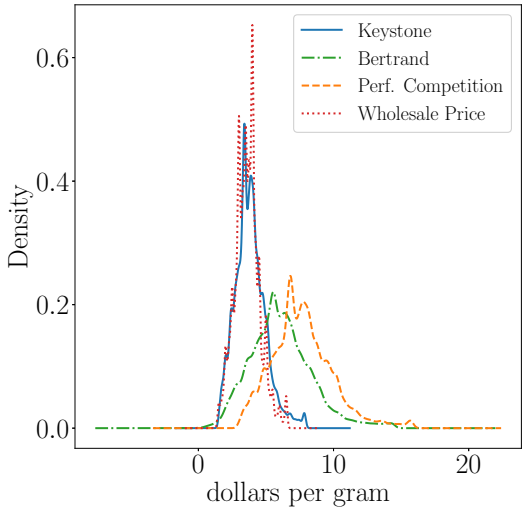
For the first two models, implied costs and markups can be immediately computed from observed prices. Given our demand estimates, we can also compute Δ_B and c_B for the Bertrand model. We specify marginal cost as the sum of observed wholesale prices p^W and additional retail costs. In turn, these include a linear function of observed shifters w and an unobserved shock. The vector w_{jt} includes product fixed effects.[33]

Inspection of Implied Costs and Markups: Before performing formal testing, we inspect model-implied costs. Given that we observe wholesale prices, and that these likely make up a large fraction of the marginal cost of selling a product, it is natural to compare the model-implied costs to these prices. Figure 2 reports implied cost distributions alongside observed wholesale prices; Keystone implies costs close to wholesale prices, while the other models do not. Table 2 confirms: the RMSE of Keystone-implied costs against wholesale prices is substantially lower, and wholesale prices represent 98% of Keystone's total marginal cost versus 49% and 65% for marginal cost pricing and Bertrand. Further inspection of markups reveals that while firms have market power under Bertrand, markups are even higher under Keystone pricing.

While not conclusive, our descriptive analysis suggests that the Keystone model is the most reasonable of the three and that the model of marginal cost pricing is implausible in this market. Additionally, our findings in Section 3.4 indicate that both Keystone and marginal cost pricing will produce the same counterfactual Laffer curve for the ad valorem tax because they have identical ad valorem tax pass-through. Thus, going forward, we focus on distinguishing Bertrand from Keystone.

[33] In Appendix F, we show the robustness of our test results to alternative specifications of cost.

FIGURE 2: Model Implied Marginal Costs



The figure plots distributions of marginal costs implied by different models of conduct, along with observed wholesale prices

TABLE 2: Descriptive Statistics of Model Implied Markups and Costs

Statistic	MC Pricing	Keystone	Bertrand
RMSE of c_m and p^W	4.38	0.78	3.02
Mean p^W/c_m	0.49	0.98	0.66
Mean Δ_m/p	0	0.50	0.22
Mean Δ_m(\$)	0	3.84	1.56
25th percentile Δ_m (\$)	0	3.10	1.19
50th percentile Δ_m (\$)	0	3.75	1.30
75th percentile Δ_m (\$)	0	4.45	1.76

The table reports descriptive statistics of model-implied markups and costs. Each column corresponds to a different model of conduct.

Implementation of the Test: To make inferences on firm conduct in finite samples, we adopt the model selection test in Rivers and Vuong (2002) (RV) introduced in Section 4, which requires instruments. Because we have variation in the data in state-wide and local ad valorem taxes, our results in Section 3.4 suggest that for an optimal ad valorem taxation counterfactual, we should use ad valorem tax instruments. However, if the researcher has a different objective, such as learning conduct for its own sake, they may want to consider different sets of relevant instruments. In addition to taxes, three other sources of variation in the data should be relevant for testing. These are (i) rival cost shifters - RC (sum of rainfall, temperature, and their lags); (ii) product characteristics - PC (own and rival number of products, total own and rival CBD and THC); (iii) wholesale prices - WP. In our setting, wholesale prices are functionally equivalent to product-specific unit tax rates.

While all of these instruments are exogenous under standard assumptions (including marginal cost pricing upstream), their relevance can be analyzed in light of our falsification framework. Assuming that the true model is one of the two candidates we test in this application, any of these instruments is plausibly relevant, i.e., capable of falsifying the wrong model, as shown in the previous examples. However, we also learn that different instruments leverage different features of the pass-through matrix to distinguish models of conduct. Thus, using each set of instruments separately to perform the test will provide robust evidence.

Beyond our falsification results, the ability of different sets of instruments to falsify conduct in a finite sample will depend on the variation present in the data. To assess the strength of instruments in the data, we report values of the F-statistic for the RV test developed in Duarte et al. (2024).

Test Results: We test the Bertrand and Keystone models using the RV test with Tax, RC, PC, and WP instruments. We report the results in Table 3.[34]

Table 3: Test Results

	Instruments:			
	(Preferred)	(Other Instruments)		
Statistic	Tax	RC	PC	WP
	(1)	(2)	(3)	(4)
T^{RV}	13.519	3.767	9.555	17.037
	$***$	$***$	$***$	$***$
F	34.338	2.126	18.653	1,639.6
	$\dagger\dagger\dagger$ $\wedge\wedge\wedge$	$\dagger\dagger\dagger$ $\wedge\wedge\wedge$	$\dagger\dagger\dagger$ $\wedge\wedge\wedge$	$\dagger\dagger\dagger$ $\wedge\wedge\wedge$

The table reports, for each set of instruments, the RV test statistics T^{RV} and the effective F-statistic (Duarte et al., 2024) for testing Bertrand versus Keystone. A positive RV test statistic suggests a better fit of Keystone. The symbol $***$ indicates rejection of the null of equal fit at the 0.01 significance level. The symbols $\dagger\dagger\dagger$ and $\wedge\wedge\wedge$ indicate that F is above the appropriate critical values for worst-case size below 0.075, and best-case power above 0.95, respectively. Both T^{RV} and the F-statistics account for two-step estimation error and clustering at the market level. $n = 187,499$.

In each column – corresponding to a set of instruments – the table reports the RV test statistic T^{RV} and the effective F-statistic (Duarte et al., 2024) for testing Bertrand versus Keystone. A positive value of the RV test statistic above 1.96 indicates rejection of the null of equal fit in favor of a better fit for Keystone at the 0.05 significance level. All sets of instruments decisively reject the null in favor of Keystone. Given that we observe rejection, we are not concerned by low power, but we may still be concerned that these results are due to weak instruments leading to size distortions. For all instruments, F is above the critical value for the worst-case size of 0.075, suggesting size distortions are likely to be minimal.

[34] Results are obtained with `pyRVtest` (Duarte et al., 2022).

Table 3 provides strong evidence that Keystone better describes conduct in this market. Across sets of instruments, each leveraging different features of the two models, the data consistently support Keystone as the superior model of conduct during our sample period, compared to Bertrand. In the next section, we explore how performing counterfactuals under the Keystone model has important implications for public policy.

5.4 Counterfactuals

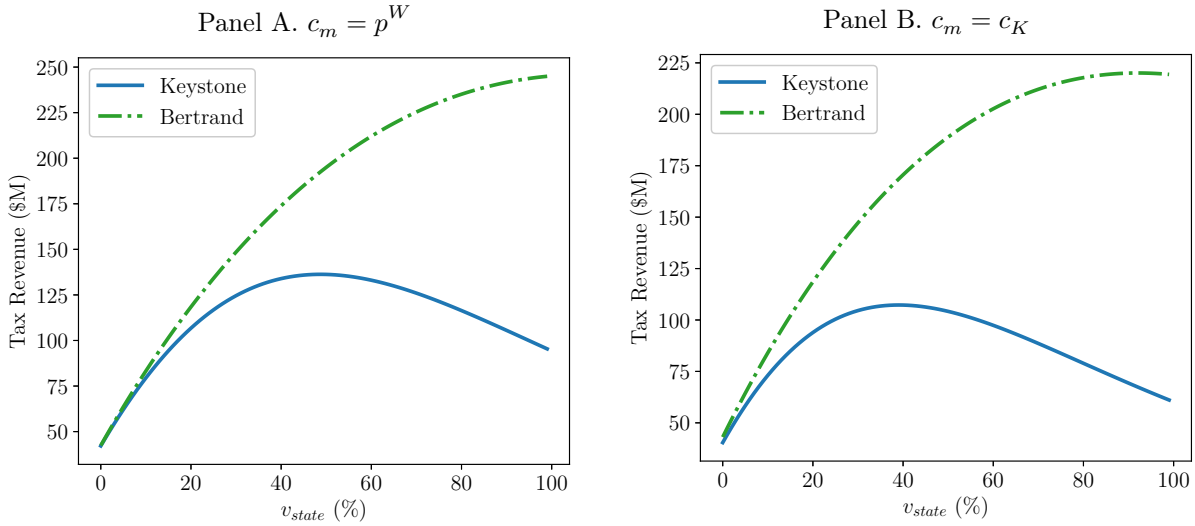
We perform three counterfactual exercises to illustrate the importance of learning conduct for optimal tax policy, focusing on the problem of setting the state-level ad valorem tax for the year 2016. While the government may have other considerations when setting taxes (e.g., curbing consumption to address externalities), we focus solely on revenue raised from the tax. Throughout, we hold market structure and local taxes fixed at their levels in the data.[35]

The first two exercises highlight the importance of learning firm conduct for optimal tax policy counterfactuals. In both, we hold the primitives of demand and cost fixed and only vary the form of conduct. As policymakers might use wholesale price as a proxy for marginal cost, the first exercise sets marginal cost equal to the observed wholesale price. The second exercise imposes the marginal cost implied under Keystone, the model that best fits the data. In both exercises, we then simulate counterfactual prices, quantities, and government revenue at different levels of the state-wide ad valorem tax for each candidate model.

Our results are plotted in Figure 3. We find that the model of conduct has substantial implications for the Laffer curve when holding demand and cost fixed, in line with the conclusions from the recent literature (Miravete et al., 2018; Hollenbeck and Uetake, 2021; O'Connell and Smith, 2024). For both formulations of cost, the results are similar: the revenue-maximizing state-level tax rate is considerably lower under Keystone than Bertrand. In fact, when using the Keystone-implied costs in Panel B, the revenue-maximizing state-level tax rate under Keystone is 36%, almost identical to the 37% rate the government adopted in 2015. The Bertrand model produces very different results; the associated Laffer curve lies well above the Keystone curve and is maximized at $v_{state} = 80\%$ (Panel B).

[35] As the purpose of our exercise is to illustrate the importance of learning conduct for deriving the Laffer curve, we also assume that a firm setting prices as a multiple of cost would continue to price at the same multiple of cost at every counterfactual tax rate.

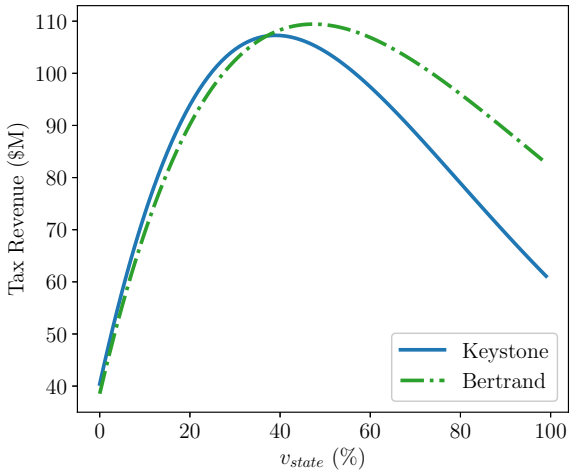
FIGURE 3: Laffer Curve in 2016 Fixing Cost Across Models



Panels A and B report, for 2016, the Laffer curve implied by different models of conduct for the state-level ad valorem tax (holding local tax rates fixed). Panel A imposes marginal cost equal to the wholesale price. Panel B imposes marginal cost equal to the estimates under Keystone pricing, the preferred model according to our testing results. We then simulate counterfactual prices, quantities, and government revenue at different levels of the tax for each candidate model.

In the third exercise, we consider the thought experiment of a policymaker who commits to a specific model of conduct to both recover costs and simulate the Laffer curve. To compare across assumptions on conduct, we obtain marginal cost estimates under each candidate model. We then use the model-specific cost estimates to simulate counterfactual prices, quantities, and government revenue at different levels of the state-wide ad valorem tax. Our results are plotted in Figure 4.

FIGURE 4: Laffer Curve in 2016 Under Model Implied Costs



For 2016, the figure reports the Laffer curve implied by different models of conduct for the state-level ad valorem tax (holding local tax rates fixed). Using marginal cost estimates obtained under each model, we simulate counterfactual prices, quantities, and government revenue at different levels of the tax.

Since market outcomes are observed in the data, the Laffer curves for Bertrand and Keystone coincide at the tax rate in the data. Compared to Bertrand, the Keystone Laffer curve is maximized at a lower state-wide ad valorem tax rate (36% vs 44%). Compared to Figure 3, allowing costs to change under each model improves the model's fit and its associated Laffer curve. However, there are quantitatively important consequences to adopting the wrong model. Setting tax at the level implied by Bertrand would lead to a loss of government revenue equal to $2.9 million.

6 Conclusion

We provide an economic characterization of what gives instruments power in conduct inference: competing models are distinguished through the causal effects of instruments on equilibrium prices, summarized by pass-through and related markup objects.

Our framework provides a formal link between the "structural" falsifiable restriction in Berry and Haile (2014) and the "reduced-form" pass-through regressions used in Sumner (1981). It also enables us to compare the relevance of various conduct instruments, including rival cost shifters, product characteristics, and tax rates. We show that different instruments target different features of the pass-through matrix.

These insights yield important practical implications for empirical work. First, researchers should use models with flexible demand curvature in order to ensure sufficient flexibility in pass-through. Second, instrument choice should be tailored to the counterfactual object of interest whenever possible, rather than guided solely by generic exclusion restrictions or ex post strength diagnostics.

We demonstrate the usefulness of our framework in an application to the Washington marijuana market, where we test conduct and compute the Laffer curve. Our conduct tests favor Keystone pricing over standard Bertrand pricing. These results are robust to various choice of instruments and demand curvature specifications, and they have substantial implications for the Laffer curve.

More broadly, the framework we develop for understanding and selecting conduct instruments is portable to other settings and policy questions. We anticipate that our results will offer a useful toolkit for researchers tackling new classes of models or instruments in a wide range of empirical settings.

References

Adams, B. and K. R. Williams (2019): "Zone Pricing in Retail Oligopoly," *American Economic Journal: Microeconomics*, 11, 124–56.

Anderson, S., A. de Palma, and B. Kreider (2001): "The Efficiency of Indirect Taxes Under Imperfect Competition," *Journal of Public Economics*, 81, 231–251.

Atalay, E., E. Frost, A. Sorensen, C. Sullivan, and W. Zhu (2025): "Scalable Demand and Markups," Working Paper.

Auerbach, A. (1985): "The Theory of Excess Burden and Optimal Taxation," in *Handbook of Public Economics*, ed. by A. Auerbach and M. Feldstein, Amsterdam: Elsevier, vol. 1, chap. 2, 61–127.

——— (2002): "Taxation and Corporate Financial Policy," in *Handbook of Public Economics*, ed. by A. Auerbach and M. Feldstein, Amsterdam: Elsevier, vol. 3, chap. 19, 1251–1292.

Backus, M., C. Conlon, and M. Sinkinson (2021): "Common Ownership and Competition in the Ready-To-Eat Cereal Industry," NBER Working Paper #28350.

Berry, S. and P. Haile (2014): "Identification in Differentiated Products Markets Using Market Level Data," *Econometrica*, 82, 1749–1797.

Berry, S., J. Levinsohn, and A. Pakes (1995): "Automobile Prices in Market Equilibrium," *Econometrica*, 63, 841–890.

Berry, S. T. and P. A. Haile (2021): "Foundations of demand estimation," in *Handbook of Industrial Organization, Volume 4*, ed. by K. Ho, A. HortaÃ§su, and A. Lizzeri, Elsevier, vol. 4 of *Handbook of Industrial Organization*, 1–62.

Birchall, C., D. Mohapatra, and F. Verboven (2024): "Estimating Substitution Patterns and Demand Curvature in Discrete-Choice Models of Product Differentiation," *The Review of Economics and Statistics*, 1–40.

Bresnahan, T. (1982): "The Oligopoly Solution Concept Is Identified," *Economics Letters*, 10, 87–92.

Brugués, F. and R. De Simone (2026): "Taxation When Markets Are Not Competitive: Evidence From a Loan Tax," Working Paper.

Bulow, J. and P. Pfleiderer (1983): "A Note on the Effect of Cost Changes on Prices," *Journal of Political Economy*, 91, 182–185.

Butters, R. A., D. Sacks, and B. Seo (2022): "How Do National Firms Respond to Local Cost Shocks?" *American Economic Review*, 112, 1737–72.

Calder-Wang, S. and G. H. Kim (2024): "Algorithmic Pricing in Multifamily Rentals: Efficiency Gains or Price Coordination?" Working Paper.

Conlon, C. and J. Gortmaker (2020): "Best Practices for Differentiated Products Demand Estimation with PyBLP," *RAND Journal of Economics*, 51, 1108–1161.

De Loecker, J. (2011): "Recovering markups from production data," *International Journal of Industrial Organization*, 29, 350–355.

De Loecker, J., J. Eeckhout, and G. Unger (2020): "The Rise of Market Power and the Macroeconomic Implications," *The Quarterly Journal of Economics*, 135, 561–644.

De Loecker, J. and P. T. Scott (2024): "Markup Estimation using Production and Demand Data: An Application to the US Brewing Industry," Working paper.

De Loecker, J. and F. Warzynski (2012): "Markups and Firm-Level Export Status," *American Economic Review*, 102, 2437–71.

DellaVigna, S. and M. Gentzkow (2019): "Uniform Pricing in U.S. Retail Chains," *Quarterly Journal of Economics*, 134, 2011–2084.

Döpper, H., A. MacKay, N. H. Miller, and J. Stiebale (2025): "Rising Markups and the Role of Consumer Preferences," *Journal of Political Economy*, 133, 2462–2505.

Duarte, M., L. Magnolfi, D. Quint, M. Sølvsten, and C. Sullivan (2026): "Conduct and Scale Economies: Evaluating Tariffs in the US Automobile Market," Working Paper.

Duarte, M., L. Magnolfi, M. Sølvsten, and C. Sullivan (2024): "Testing Firm Conduct," *Quantitative Economics*, 15, 571–606.

Duarte, M., L. Magnolfi, M. Sølvsten, C. Sullivan, and A. Tarascina (2022): "`pyRVtest`: A Python Package for Testing Firm Conduct," https://github.com/anyatarascina/pyRVtest.

Escudero, M. (2018): "Rule-of-Thumb Pricing: Retail Cannabis in Washington State," Working Paper.

Goeree, M. S. (2008): "Limited Information and Advertising in the U.S. Personal Computer Industry," *Econometrica*, 76, 1017–1074.

Grieco, P. L. E., C. Murry, and A. Yurukoglu (2023): "The Evolution of Market Power in the U.S. Automobile Industry," *The Quarterly Journal of Economics*, 139, 1201–1253.

Grigolon, L. and F. Verboven (2014): "Nested Logit or Random Coefficients Logit? A Comparison of Alternative Discrete Choice Models of Product Differentiation," *Review of Economics and Statistics*, 96, 916–935.

Hollenbeck, B., S. Hristakeva, and K. Uetake (2026): "Retailer Price Competition and Assortment Differentiation: Evidence from Entry Lotteries," Working Paper.

Hollenbeck, B. and K. Uetake (2021): "Taxation and Market Power in the Legal Marijuana Industry," *RAND Journal of Economics*, 52, 559–595.

Jaffe, S. and E. G. Weyl (2013): "The First-order Approach to Merger Analysis," *American Economic Journal: Microeconomics*, 5, 188–218.

Kroft, K., J.-W. Laliberté, R. Leal-Vizcaíno, and M. Notowidigdo (2023): "Salience and Taxation with Imperfect Competition," *Review of Economic Studies*, 91, 403–437.

Lau, L. (1982): "On Identifying the Degree of Competitiveness from Industry Price and Output Data," *Economics Letters*, 10, 93–99.

MacKay, A., N. Miller, M. Remer, and G. Sheu (2014): "Bias in Reduced-form Estimates of Pass-through," *Economics Letters*, 123, 200–202.

Magnolfi, L., D. Quint, C. Sullivan, and S. Waldfogel (2022): "Falsifying Models of Firm Conduct," Working Paper.

Miller, N., G. Sheu, and M. Weinberg (2021): "Oligopolistic Price Leadership and Mergers: The United States Beer Industry," *American Economic Review*, 111, 3123–59.

Miller, N. and M. Weinberg (2017): "Understanding the Price Effects of the MillerCoors Joint Venture," *Econometrica*, 85, 1763–1791.

Miravete, E., K. Seim, and J. Thurk (2018): "Market Power and the Laffer Curve," *Econometrica*, 86, 1651–1687.

——— (2020): "One Markup to Rule Them All: Taxation by Liquor Pricing Regulation," *American Economic Journal: Microeconomics*, 12, 1–41.

——— (2023): "Pass-through and tax incidence in differentiated product markets," *International Journal of Industrial Organization*, 90, 102985.

——— (2026): "Elasticity and Curvature of Discrete Choice Demand Models," Working Paper.

O'Connell, M. and K. Smith (2024): "Optimal Sin Taxation and Market Power," *American Economic Journal: Applied Economics*, forthcoming.

Ramsey, F. (1927): "A Contribution to the Theory of Taxation," *The Economic Journal*, 37, 47–61.

Rivers, D. and Q. Vuong (2002): "Model Selection Tests for Nonlinear Dynamic Models," *Econometrics Journal*, 5, 1–39.

Roussille, N. and B. Scuderi (2026): "Bidding for Talent: A Test of Conduct in a High-Wage Labor Market," Working Paper.

Sharma, G. (2025): "Collusion Among Employers in India," Working Paper.

Slemrod, J. (1990): "Optimal Taxation and Optimal Tax Systems," *Journal of Economic Perspectives*, 4, 157–178.

Starc, A. and T. G. Wollmann (2025): "Does entry remedy collusion? Evidence from the generic prescription drug cartel," *American Economic Review*, 115, 1400–1438.

Suits, D. and R. Musgrave (1953): "Ad Valorem and Unit Taxes Compared," *The Quarterly Journal of Economics*, 67, 598–604.

Sumner, D. (1981): "Measurement of Monopoly Behavior: an Application to the Cigarette Industry," *Journal of Political Economy*, 89, 1010–1019.

Villas-Boas, S. (2007): "Vertical Relationships between Manufacturers and Retailers: Inference with Limited Data," *Review of Economic Studies*, 74, 625–652.

Weyl, G. and M. Fabinger (2013): "Pass-Through as an Economic Tool: Principles of Incidence under Imperfect Competition," *Journal of Political Economy*, 121, 528–583.

Online Appendix

Appendix A Proofs

Proof of Lemma 1. As we note in the text, in our parametric framework, the falsifiable restriction in Equation (28) of Berry and Haile (2014) is[36]

$$E[\omega_{mjt} \mid \mathrm{w}_{jt}, z_{jt}] \quad = \quad E[p_{jt} - \Delta_{mjt} - \bar{c}_{mj}(\mathrm{w}_{jt}) \mid \mathrm{w}_{jt}, z_{jt}] \quad = \quad 0 \qquad a.s.$$

Since observed prices are generated under the true model as

$$p_{jt} \quad = \quad \Delta_{0jt} + c_{0jt} \quad = \quad \Delta_{0jt} + \bar{c}_{0j}(\mathrm{w}_{jt}) + \omega_{0jt}$$

and $E[\omega_{0jt} \mid \mathrm{w}_{jt}, z_{jt}] = 0$ under Assumption 3, the falsifiable restriction is equivalent to

$$E[\Delta_{0jt} + \bar{c}_{0j}(\mathrm{w}_{jt}) + \omega_{0jt} - \Delta_{mjt} - \bar{c}_{mj}(\mathrm{w}_{jt}) \mid \mathrm{w}_{jt}, z_{jt}] \quad = \quad 0 \qquad a.s.$$

or equivalently

$$E[\Delta_{0jt} - \Delta_{mjt} \mid \mathrm{w}_{jt}, z_{jt}] \quad = \quad \bar{c}_{mj}(\mathrm{w}_{jt}) - \bar{c}_{0j}(\mathrm{w}_{jt}) \qquad a.s.$$

giving the result. □

Proof of Lemma 2. We prove the inverse of both directions. First, suppose the model is not falsified. Assumption 3 requires $\frac{\partial \bar{c}_{mj}}{\partial z_{jt}} = \frac{\partial \bar{c}_{0j}}{\partial z_{jt}}$, so changes in the instruments do not change $\bar{c}_{mjt}\left(\mathrm{w}_{jt}\right) - \bar{c}_{0jt}\left(\mathrm{w}_{jt}\right)$. This means that falsification depends on the impact of the instruments on the conditional expectation of $\Delta_{0jt} - \Delta_{mjt}$. For each $z_{jt}^{(k)}$, the limit of this difference is

$$\lim_{h \to 0} \frac{E[\Delta_{0jt} - \Delta_{mjt} \mid \mathrm{w}_{jt}, z_{jt} = \tilde{z}_{jt} + h_k] - E[\Delta_{0jt} - \Delta_{mjt} \mid \mathrm{w}_{jt}, z_{jt} = \tilde{z}_{jt}]}{h}$$

where h_k is a K-vector of zeros with the scalar h in the k-th position. The marginal effect of the k-th instrument $z_{jt}^{(k)}$ on the conditional expectation of $\Delta_{0jt} - \Delta_{mjt}$ is the average difference in the marginal effect of $z_{jt}^{(k)}$ on Δ_{0jt} and Δ_{mjt}. Furthermore, the model is not falsified, so the expectation $E[\Delta_{0jt} - \Delta_{mjt} | \mathrm{w}_{jt}, z_{jt}] = 0$ almost everywhere and is therefore constant with respect to z_{jt} almost everywhere. Altogether, this implies the following:

[36] See Section 6, Case 2 in Berry and Haile (2014) for a discussion of their non-parametric environment.

$$E\left[\frac{\mathrm{d}\Delta_{0jt}}{\mathrm{d}z_{jt}^{(k)}} - \frac{\mathrm{d}\Delta_{mjt}}{\mathrm{d}z_{jt}^{(k)}} \mid \mathrm{w}_{jt}, z_{jt}\right] = 0 \qquad a.s.$$

We can do this for each $k = 1, \ldots, K$, giving the result.

For the opposite direction, if $E\left[\frac{\mathrm{d}\Delta_{0jt}}{\mathrm{d}z_{jt}} - \frac{\mathrm{d}\Delta_{mjt}}{\mathrm{d}z_{jt}^{(k)}} \mid \mathrm{w}_{jt}, z_{jt}\right] = 0$ almost surely, then $E[\Delta_{0jt} - \Delta_{mjt} \mid \mathrm{w}_{jt}, z_{jt}]$ must be the same for almost all values of z_{jt}. If so, define

$$\bar{c}_{mj}(\mathrm{w}_{jt}) \quad = \quad \bar{c}_{0j}(\mathrm{w}_{jt}) + E_{z_{jt}}\left[E[\Delta_{0jt} - \Delta_{mjt} \mid \mathrm{w}_{jt}, z_{jt}]\right]$$

and $\bar{c}_{mj}$ satisfies the equality condition in Lemma 1 almost surely, so the model is not falsified. □

Proof of Lemma 3. First, we note that $\frac{\mathrm{d}\Delta_{0jt}}{\mathrm{d}z_{jt}} - \frac{\mathrm{d}\Delta_{mjt}}{\mathrm{d}z_{jt}}$ is the j-th row of $\frac{\mathrm{d}\Delta_{0t}}{\mathrm{d}z_{jt}} - \frac{\mathrm{d}\Delta_{mt}}{\mathrm{d}z_{jt}}$. Working with the latter term and breaking it up into the direct and indirect effects as in the main text, we have

$$\begin{aligned}
\frac{\mathrm{d}\Delta_{0t}}{\mathrm{d}z_{jt}} - \frac{\mathrm{d}\Delta_{mt}}{\mathrm{d}z_{jt}} &= \frac{\partial\Delta_{0t}}{\partial z_{jt}} - \frac{\partial\Delta_{mt}}{\partial z_{jt}} + \left(\frac{\partial\Delta_{0t}}{\partial p_t} - \frac{\partial\Delta_{mt}}{\partial p_t}\right)\frac{\mathrm{d}p_0}{\mathrm{d}z_{jt}} \\
&= \frac{\partial\Delta_{0t}}{\partial z_{jt}} - \frac{\partial\Delta_{mt}}{\partial z_{jt}} + \left(P_{mt}^{-1} - P_{0t}^{-1}\right)P_{0t}\left(\frac{\partial\Delta_{0t}}{\partial z_{jt}} + \frac{\partial\bar{c}_{0t}}{\partial z_{jt}}\right) \\
&= -\frac{\partial\Delta_{mt}}{\partial z_{jt}} + P_{mt}^{-1}P_{0t}\left(\frac{\partial\Delta_{0t}}{\partial z_{jt}} + \frac{\partial\bar{c}_{0t}}{\partial z_{jt}}\right) - \frac{\partial\bar{c}_{0t}}{\partial z_{jt}} \\
&= -\frac{\partial\Delta_{mt}}{\partial z_{jt}} + P_{mt}^{-1}P_{0t}\left(\frac{\partial\Delta_{0t}}{\partial z_{jt}} + \frac{\partial\bar{c}_{0t}}{\partial z_{jt}}\right) - \frac{\partial\bar{c}_{0t}}{\partial z_{jt}} + \frac{\partial\bar{c}_{mt}}{\partial z_{jt}} - \frac{\partial\bar{c}_{mt}}{\partial z_{jt}} \\
&= P_{mt}^{-1}\left(P_{0t}\left(\frac{\partial\Delta_{0t}}{\partial z_{jt}} + \frac{\partial\bar{c}_{0t}}{\partial z_{jt}}\right) - P_{mt}\left(\frac{\partial\Delta_{mt}}{\partial z_{jt}} + \frac{\partial\bar{c}_{mt}}{\partial z_{jt}}\right)\right) + \frac{\partial\bar{c}_{mt}}{\partial z_{jt}} - \frac{\partial\bar{c}_{0t}}{\partial z_{jt}} \\
&= P_{mt}^{-1}\left(\frac{\mathrm{d}p_{0t}}{\mathrm{d}z_{jt}} - \frac{\mathrm{d}p_{mt}}{\mathrm{d}z_{jt}}\right) + \frac{\partial\bar{c}_{mt}}{\partial z_{jt}} - \frac{\partial\bar{c}_{0t}}{\partial z_{jt}}.
\end{aligned}$$

Taking the j-th row of the final expression yields the statement in the lemma because $\frac{\partial\bar{c}_{mjt}}{\partial z_{jt}} - \frac{\partial\bar{c}_{0jt}}{\partial z_{jt}} = 0$ by Assumption 3.

□

Proof of Proposition 1. Follows immediately Lemmas 2 and 3.

□

Proof of Proposition 2. Given (4), we have (i) $c_{0jt} = \bar{c}_{0j}(\mathrm{w}_{jt}) + \omega_{0jt} = \frac{1}{\nu_t}p_{jt} - \frac{1}{\nu_t}\Delta_{0jt} - \tau_t$;

and (ii) $c_{mjt} = \bar{c}_{mj}(\mathrm{w}_{jt}) + \omega_{mjt} = \frac{1}{\nu_t} p_{jt} - \frac{1}{\nu_t}\Delta_{mjt} - \tau_t$. So,

$$\begin{aligned}
\omega_{mjt} &= \tfrac{1}{\nu_t} p_{jt} - \tfrac{1}{\nu_t}\Delta_{mjt} - \tau_t - \bar{c}_{mj}(\mathrm{w}_{jt}) \\
&= \tfrac{1}{\nu_t}\left(\Delta_{0jt} - \Delta_{mjt}\right) + \bar{c}_{0j}(\mathrm{w}_{jt}) - \bar{c}_{mj}(\mathrm{w}_{jt}) + \omega_{0jt}
\end{aligned}$$

Since $\bar{c}_{0j}\left(\mathrm{w}_{jt}\right)$, $\bar{c}_{mj}\left(\mathrm{w}_{jt}\right)$, and $E\left[\omega_{0jt}|\mathrm{w}_{jt}, z_t\right] = 0$ don't depend on the tax instruments, falsification occurs if $E\left[\frac{1}{\nu_t}\left(\Delta_{0jt} - \Delta_{mjt}\right) \mid \mathrm{w}_{jt}, z_{jt}\right] \neq 0$ w.p.p. for some j.

The mechanics driving Lemma 2 immediately extend so that falsification occurs if and only if $E\left[\frac{\mathrm{d}}{\mathrm{d}z_{jt}}\left\{\frac{1}{\nu_t}\left(\Delta_{0jt} - \Delta_{mjt}\right)\right\} \mid \mathrm{w}_{jt}, z_{jt}\right] \neq 0$ w.p.p. To complete the proof, we must show that the derivative inside the expectation takes the form in the proposition.

Taxes do not directly enter the markups; they only affect markups indirectly through the observed prices, which are determined by the true model p_{0t}. So, we have $\frac{\mathrm{d}\Delta_{mt}}{\mathrm{d}z_{jt}} = \left(I - P_{mt}^{-1}\right)\frac{\mathrm{d}p_{0t}}{\mathrm{d}z_{jt}}$ for any model m (including $m = 0$). Differentiating $\frac{1}{\nu_t}\left(\Delta_{0t} - \Delta_{mt}\right)$ w.r.t. the unit tax yields

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}\tau_t}\left\{\frac{1}{\nu_t}\left(\Delta_{0t} - \Delta_{mt}\right)\right\} &= \frac{1}{\nu_t}\left(\left(I - P_{0t}^{-1}\right)\frac{\mathrm{d}p_{0t}}{\mathrm{d}\tau_t} - \left(I - P_{mt}^{-1}\right)\frac{\mathrm{d}p_{0t}}{\mathrm{d}\tau_t}\right) \\
&= \frac{1}{\nu_t}\left(P_{mt}^{-1}\frac{\mathrm{d}p_{0t}}{\mathrm{d}\tau_t} - P_{0t}^{-1}\frac{\mathrm{d}p_{0t}}{\mathrm{d}\tau_t}\right) \\
&= \frac{1}{\nu_t}\left(P_{mt}^{-1}P_{0t}\iota - P_{0t}^{-1}P_{0t}\iota\right) \\
&= \frac{1}{\nu_t}P_{mt}^{-1}\left(P_{0t}\iota - P_{mt}\iota\right) \\
&= \frac{1}{\nu_t}P_{mt}^{-1}\left(\frac{\mathrm{d}p_{0t}}{\mathrm{d}\tau_t} - \frac{\mathrm{d}p_{mt}}{\mathrm{d}\tau_t}\right)
\end{aligned}$$

where the third-line and final-line equalities arise because $\frac{\mathrm{d}p_{0t}}{\mathrm{d}\tau_t} = P_{0t}\iota$ and $\frac{\mathrm{d}p_{mt}}{\mathrm{d}\tau_t} = P_{mt}\iota$, as shown in Section 3.3.

Differentiating w.r.t. the ad valorem tax yields

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}\nu_t}\left\{\frac{1}{\nu_t}\left(\Delta_{0t}-\Delta_{mt}\right)\right\} &= -\frac{1}{\nu_t^2}\left(\Delta_{0t}-\Delta_{mt}\right)+\frac{1}{\nu_t}\left(\left(I-P_{0t}^{-1}\right)\frac{\mathrm{d}p_{0t}}{\mathrm{d}\nu_t}-\left(I-P_{mt}^{-1}\right)\frac{\mathrm{d}p_{0t}}{\mathrm{d}\nu_t}\right)\\
&= -\frac{1}{\nu_t^2}\left(\Delta_{0t}-\Delta_{mt}\right)+\frac{1}{\nu_t}\left(P_{mt}^{-1}\frac{\mathrm{d}p_{0t}}{\mathrm{d}\nu_t}-P_{0t}^{-1}\frac{\mathrm{d}p_{0t}}{\mathrm{d}\nu_t}\right)\\
&= \frac{1}{\nu_t^2}\left(P_{mt}^{-1}P_{0t}\left(p_t-\Delta_{0t}\right)-P_{0t}^{-1}P_{0t}\left(p_t-\Delta_{0t}\right)-\Delta_{0t}+\Delta_{mt}\right)\\
&= \frac{1}{\nu_t^2}\left(P_{mt}^{-1}P_{0t}\left(p_t-\Delta_{0t}\right)-p_{0t}+\Delta_{mt}\right)\\
&= \frac{1}{\nu_t^2}P_{mt}^{-1}\left(P_{0t}\left(p_t-\Delta_{0t}\right)-P_{mt}\left(p_t-\Delta_{0t}\right)\right)\\
&= \frac{1}{\nu_t}\left(\frac{\mathrm{d}p_{0t}}{\mathrm{d}\nu_t}-\frac{\mathrm{d}p_{mt}}{\mathrm{d}\nu_t}\right)
\end{aligned}$$

where the third-line and final-line equality arise because $\frac{\mathrm{d}p_{0t}}{\mathrm{d}\nu_t}=\frac{1}{\nu_t}P_{0t}\left(p_t-\Delta_{0t}\right)$ and $\frac{\mathrm{d}p_{mt}}{\mathrm{d}\nu_t}=\frac{1}{\nu_t}P_{mt}\left(p_t-\Delta_{mt}\right)$, as shown in Section 3.3.

Taking the j'th row both cases, we obtain $\frac{\mathrm{d}}{\mathrm{d}z_{jt}}\left\{\frac{1}{\nu_t}\left(\Delta_{0jt}-\Delta_{mjt}\right)\right\}=\frac{1}{\nu_t}\left(\frac{\mathrm{d}p_{0jt}}{\mathrm{d}z_{jt}}-\frac{\mathrm{d}p_{mjt}}{\mathrm{d}z_{jt}}\right)$ for both tax rate instruments. This completes the proof. □

Appendix B Connection to Pass-Through Regressions

Early tests of market power (e.g., Sumner, 1981) relied on reduced-form measures of cost (or, equivalently, unit-tax) pass-through. Our discussion of pass-through matrices as a fundamental observable implication of models of conduct developed in Sections 2 and 3 suggests that there is intuitive merit to this idea, but also that there are limitations. Pass-through rates P_t typically vary with prices and other market characteristics and cannot be summarized by low-dimensional structural parameters except in a limited class of conduct models. For these special cases, however, we can show that the pass-through regression approach used in the early literature emerges as a special case of our more general framework.

Consider the class of "rule-of-thumb" pricing models where firms set $p_{jt} = \phi_0 c_{0jt}$ for some fixed ϕ_0. Equivalently, $p_{jt} = \frac{\phi_0 - 1}{\phi_0} p_{jt} + c_{0jt}$, so this form of pricing satisfies Assumption 1 with $\Delta_0(p_t, x_t, \xi_t) = \frac{\phi_0 - 1}{\phi_0} p_t$. Marginal cost pricing ($\phi_m = 1$) and Keystone pricing ($\phi_m = 2$) are special cases in this rule-of-thumb class. In this class of models, own cost pass-through equals ϕ_m and rival cost pass-through is zero.

For simplicity, assume marginal cost is linear in w_{jt}, so that $c_{0jt} = \mathrm{w}_{jt}\gamma + \omega_{0jt}$, and that the data includes a unit tax τ_t which varies across markets. If data are generated by a model in the rule-of-thumb class, then cost pass-through can be measured via linear regression with own price p_{jt} as the dependent variable and the unit tax τ_t as the independent variable, after residualizing them w.r.t. the other own-cost shifters, w_{jt} (Sumner, 1981).[37] The coefficient on the unit tax in this pass-through regression (on residualized variables), $\beta_{PTR} \equiv \mathrm{cov}(\tau_t, p_{jt})/\mathrm{var}(\tau_t)$, equals ϕ_0. Hence, any rule-of-thumb model where $\phi_m \neq \beta_{PTR}$ is falsified. For example, if $\beta_{PTR} \neq 1$, then marginal cost pricing is falsified.

The falsification framework with instrumental variables nests this simple pass-through regression approach as a special case. To see this, assume the true model is in the rule-of-thumb class, and suppose we use the unit tax τ_t as the instrument – with all variables residualized w.r.t. w_{jt}. The falsifiable restriction can be implemented through a covariance restriction: $E\left[\omega_{0jt}\tau_t\right] = 0$. With p_{jt} and τ_t residualized w.r.t. w_{jt}, we then have

$$\begin{aligned} E\left[\omega_{mjt}\tau_t\right] &= E\left[\left(p_{jt} - \frac{\phi_m - 1}{\phi_m} p_{jt} - \tau_t - \mathrm{w}_{jt}\gamma\right)\tau_t\right] \\ &= E\left[p_{jt}\tau_t\right] - \frac{\phi_m - 1}{\phi_m} E\left[p_{jt}\tau_t\right] - E\left[\tau_t^2\right] \\ &= \frac{1}{\phi_m}\mathrm{cov}(\tau_t, p_{jt}) - \mathrm{var}(\tau_t) \end{aligned}$$

[37] The Frisch-Waugh-Lovell theorem establishes that residualizing w.r.t. the dependent and independent variables w.r.t. w_{jt} is equivalent to controlling for w_{jt} by including it in the (non-residualized) regression.

for rule-of-thumb model m.[38] Falsification requires $E\left[\omega_{mjt}\tau_t\right] \neq 0$, which occurs if and only if $\phi_m \neq \text{cov}(\tau_t, p_{jt})/\text{var}(\tau_t)$. That is, falsification occurs when $\phi_m \neq \beta_{PTR}$, so falsifying a rule-of-thumb model with unit tax instruments amounts to comparing ϕ_m to the coefficient of the pass-through regression. Thus, the instrument-based approach coincides with the pass-through regression approach in this particular setting.

Despite its intuitive appeal, the pass-through regression approach to conduct inference has several conceptual and practical limitations. First, the required variation in taxes or observed cost components may not be available in the data. Second, outside of constant pass-through environments, such as the one described above or the homogeneous product environments in Bulow and Pfleiderer (1983), own and rival cost pass-through are not structural parameters. Instead, in market t, cost pass-through under model m is represented by the $J \times J$ matrix P_{mt}, which varies with realizations of both observable and unobservable determinants of demand and cost. Consequently, it is generally not possible to directly recover the elements of P_{mt} via regression (MacKay et al., 2014).

The IV approach to falsification generalizes the pass-through approach in a way that avoids these econometric issues and allows for a much wider variety of conduct inference. Our analysis in Sections 2 and 3 reveals that the link between the IV approach and the pass-through regression approach runs much deeper than this simple mechanical connection. Proposition 1 establishes a key role for the pass-through matrix in IV-based inference on conduct. Furthermore, our analysis in Section 3 reveals that several instruments distinguish conduct models solely through the pass-through matrix. In short, the economic intuition underlying the pass-through regression approach plays a crucial role in conduct inference, even when the basic pass-through regression itself is invalid.

[38] Note that residualized variables have zero mean, so for two variables x and y, $E[xy] = \text{cov}(x, y)$ and $E[x^2] = \text{var}(x)$.

Appendix C Connecting Falsification and Point Identification when Models are Nested

The results in Sections 2 and 3 are cast in terms of falsification of a candidate model m, and we pursue a testing approach in our simulations and application, but the results in the paper are also useful to inform identification (and thus estimation) exercises. In particular, our results can guide ex-ante instrument selection to avoid irrelevant instruments for estimation.

Similar to Magnolfi and Sullivan (2022), consider a setting where different candidate models belong to a parametric class, so that markups can be written as $\Delta(\theta)$ for a vector of parameters $\theta \in \Theta$. In this class of models, falsification proceeds as in our general discussion in Sections 2 and 3. In particular, a model m corresponding to a parameter value θ_m, is falsified by instruments z_{jt} under the general conditions laid out in Proposition 1. Within the parametric class of nested models, however, we can also discuss the identification of the true model, characterized by θ_0. For any value of θ, let the implied cost be $p_{jt} - \Delta_{jt}(\theta) = c(\mathrm{w}_{jt};\theta) = \bar{c}(\mathrm{w}_{jt};\theta) + \omega_{jt}(\theta)$. We say that the true model is *point identified by the instruments* z_{jt} when $E[\omega_{jt}(\theta)|\mathrm{w}_{jt}, z_{jt}] = 0$ if and only if $\theta = \theta_0$. Thus, point identification requires that all models m for which $\theta_m \neq \theta_0$ are falsified by the instruments z_{jt}. Conversely, when there exist a model m such that $\theta_m \neq \theta_0$ which is not falsified by z_{jt}, θ_0 is not point identified by these instruments. Clearly, instruments that ensure falsification for all incorrect models in the class are prime candidates to be used for identification and estimation. However, instruments for which an incorrect model is not falsified would lead to a failure of point identification.

As an example of a conduct model defined by a continuous parameter, IO economists often use profit weights as a reduced-form way to model collusion or common ownership (e.g., Backus et al. (2021)). We illustrate how our framework for understanding falsification through pass-through can be useful for ex-ante instrument selection for identification and estimation with two examples based on such profit weights.

Example 10: We remain in the example environment of the paper – two single-product firms and logit demand. Suppose the two firms compete in quantities a la Cournot, but instead of maximizing its own profit, each firm maximizes a weighted sum of its own and the other firm's profits. Specifically, each firm j chooses quantity s_{jt} to solve

$$\max_{s_{jt}} \left\{ s_{jt}(p_{jt}(\cdot) - c_{jt}) + \theta_j s_{-jt}(p_{-jt}(\cdot) - c_{-jt}) \right\}$$

where $-j$ refers to the identity of the rival firm. This nests standard Cournot competition

(when $\theta_1 = \theta_2 = 0$) and perfect collusion/joint profit maximization (when $\theta_1 = \theta_2 = 1$), along with intermediate cases that might be interpreted as "imperfect collusion". The corresponding markups under logit demand are

$$\Delta_{WCt}(\theta) \;=\; \begin{bmatrix} \frac{1-(1-\theta_1)s_{2t}}{\alpha s_{0t}} \\ \frac{1-(1-\theta_2)s_{1t}}{\alpha s_{0t}} \end{bmatrix},$$

with WC standing for the Weighted Cournot model.

For this setting, our falsification framework can be used to show that either cost or demand side instruments allow point identification of the profit weights $\theta = (\theta_1, \theta_2)$:

Result 1. *If the data are generated by Cournot competition with profit weight equal to $\theta = \theta_0 \in [0,1]^2$, then either cost side or demand side instruments point identify the true parameter.*

To see this result, we first simplify the first-order conditions and work out the pass-through and inverse pass-through matrices

$$P_{WCt}^{-1} \;=\; \frac{1}{s_{0t}} \begin{bmatrix} 1-s_{2t} & \theta_1 s_{2t} \\ \theta_2 s_{1t} & 1-s_{1t} \end{bmatrix} \qquad \text{and} \qquad P_{WCt} \;=\; \frac{s_{0t}}{\kappa_{WCt}} \begin{bmatrix} 1-s_{1t} & -\theta_1 s_{2t} \\ -\theta_2 s_{1t} & 1-s_{2t} \end{bmatrix}$$

where $\kappa_{WCt} = (1-s_{1t})(1-s_{2t}) - \theta_1\theta_2 s_{1t}s_{2t}$.

Suppose the true model is weighted Cournot competition with weights $\theta_0 = (\theta_{01}, \theta_{02})$, and we are interested in falsifying a model of weighted Cournot with misspecified weights $\theta_m = (\theta_{m1}, \theta_{m2}) \neq \theta_0$. Focusing on the off-diagonal terms and dropping constants, we can calculate

$$P_{WCmt}^{-1} P_{WC0t} \quad \propto \quad \begin{bmatrix} \star & (\theta_{m1}-\theta_{01})s_{2t}(1-s_{2t}) \\ (\theta_{m2}-\theta_{02})s_{1t}(1-s_{1t}) & \star \end{bmatrix}$$

Thus, if $\theta_{m1} > \theta_{01}$, the top-right off-diagonal is always positive, and therefore positive when its expectation is taken over unobservables; if $\theta_{m1} < \theta_{01}$, it is always negative, hence negative in expectation. Likewise, if $\theta_{m2} > \theta_{02}$, the bottom-left term is always positive, and if $\theta_{m2} < \theta_{02}$ always negative. Thus, if $\theta_m \neq \theta_0$, the matrix $[P_m^{-1}P_0]^*$ is not diagonal, so Remark 1 implies that incorrect model within the class is falsified by cost side instruments. As for demand-side instruments, the profit-weighted Cournot model with logit demand satisfies the index restriction in Remark 4, implying that demand-side instruments have the same falsifying content as cost shifter instruments here. So, demand-side instruments also falsify any incorrect model. As a result, either type of instrument allows for point identification of θ_0. •

Example 11: In the same environment, suppose now that the two firms compete in prices

with profit weights. For simplicity, suppose the two firms' profit weights are the same, so each firm j chooses price p_{jt} to solve

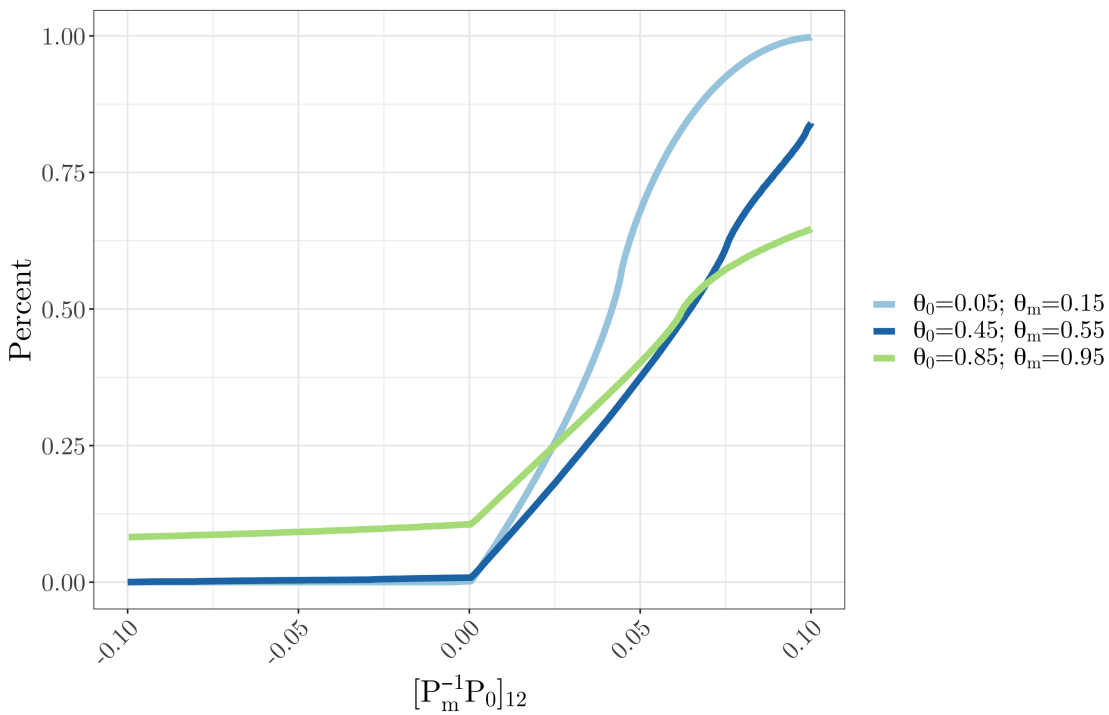
$$\max_{p_{jt}} \left\{ (p_{jt} - c_{jt}) s_{jt}(p_t) + \theta (p_{-jt} - c_{-jt}) s_{-jt}(p_t) \right\}$$

Under logit demand, the two firms' first-order conditions give the markup function

$$\Delta_{WBt}(\theta) \quad = \quad \begin{bmatrix} \frac{1-(1-\theta)s_{2t}}{\alpha s_{0t} + \alpha(1-\theta^2) s_{1t} s_{2t}} \\ \\ \frac{1-(1-\theta)s_{1t}}{\alpha s_{0t} + \alpha(1-\theta^2) s_{1t} s_{2t}} \end{bmatrix}$$

with WB standing for the Weighted Bertrand model.

FIGURE 5: CDF of $[P_{mt}^{-1} P_{0t}]_{12}$ for Bertrand Profit Weight Models



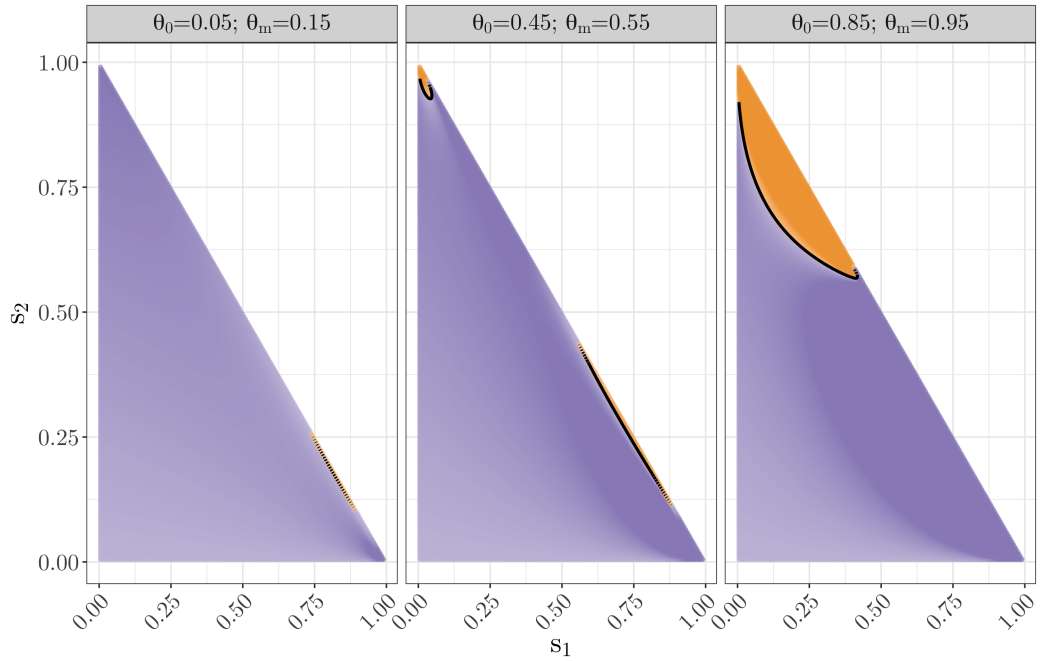
This figure plots distributions of $[P_{mt}^{-1} P_{0t}]_{12}$ over realizations of (s_{1t}, s_{2t}) for three different values of (θ_0, θ_m).

While we obtain identification in Weighted Cournot, irrespective of the value of θ_0 and of other observables, in the case of Weighted Bertrand, results are more nuanced. For a given true value of θ_0 and a given misspecified model $\theta_m \neq \theta_0$, the off-diagonal terms of the matrix $P_m^{-1} P_0$ can be either positive or negative, but are not typically zero; and the diagonal terms can be greater than or less than 1, but are not typically equal to either 1 or to each other. This means there are realizations of observables (w_t, z_t) for which $[P_m^{-1} P_0]^\star$ could in principle

be diagonal, but it will not typically be – for example, because some off-diagonal element of $P_{mt}^{-1}P_{0t}$ varies continuously and could take either sign, so its expectation being exactly zero requires a non-generic distribution of unobservables. Still, falsification could in principle fail, and therefore, identification could fail as well.

To illustrate this possibility, we compute the top-right off-diagonal term, $[P_m^{-1}P_0]_{12}$, for a large sample of uniform draws of (s_{1t}, s_{2t}), and display the resulting distribution in Figure 5 for three representative pairs of true and misspecified models (θ_0, θ_m). While the distributions all have support that includes 0, the distributions are continuous, so getting a conditional expectation $E([P_{mt}^{-1}P_{0t}]_{12}) = 0$ would only hold for a non-generic set of observables.

FIGURE 6: Values of $[P_{mt}^{-1}P_{0t}]_{12}$ for Realizations of Market Shares in Bertrand Profit Weight Models



This figure plots magnitudes of $[P_{mt}^{-1}P_{0t}]_{12}$ over the simplex of (s_{1t}, s_{2t}) for three different values of (θ_0, θ_m). Darker purple (orange) shading indicates more positive (negative) values. Black indicates a zero value.

We can also examine the value of the off-diagonal term at each point in the simplex of realizations of market shares (s_{1t}, s_{2t}). In each panel of Figure 6, we do so for three pairs of θ_0 and θ_m such that $\theta_m - \theta_0 = 0.1$. For each realization on the simplex, we indicate the value of $[P_m^{-1}P_0]_{12}$ with color gradients – darker orange indicates negative values that are larger in magnitude and darker purple indicates positive values that are larger in magnitude. The points where $[P_m^{-1}P_0]_{12} = 0$ are indicated in black. Immediately, one sees that for the vast majority of realizations of shares, $[P_m^{-1}P_0]_{12} > 0$. In fact, $[P_m^{-1}P_0]_{12}$ is always positive whenever the market share of both products is below 0.6. For an empirically relevant example, take the setting in Miller and Weinberg (2017). There, the outside option is defined

in such a way that the market share of any product is less than 0.5 in all markets. In that case, the average of $[P_m^{-1}P_0]_{12}$ is positive in all three panels, and falsification of the wrong profit weight is possible with cost or demand side instruments.

As in the previous example, these conduct models with logit demand satisfy the linear index restriction in Remark 4, so $[P_m^{-1}P_0]^*$ being non-diagonal suffices for falsification of the wrong model with either cost- or demand-side instruments. With variation in observables, point identification seems virtually guaranteed, as the knife-edge result of positive and negative values of $[P_m^{-1}P_0]_{12}$ canceling out in expectation seems impossible across different realizations of observables. •

Appendix D Monte Carlo Simulations

We illustrate the main results of the paper with simulations tied to the examples described in the preceding sections. To align the environment with the falsification framework, we simulate data for 50,000 markets using the simulation class in `PyBLP` (Conlon and Gortmaker, 2020). In each market t, the number of products J_t is a randomly chosen integer between two to ten, leaving us with 319,719 observations in the sample. Each product j is produced by a single product firm.

For demand, we adopt a simple logit framework, in line with our falsification examples. Consumer i gets indirect utility of consuming product j in market t given by:

$$u_{ijt} = x_{jt}\beta + \alpha p_{jt} + \xi_{jt} + \epsilon_{ijt}$$

where x_{jt} is a vector containing a constant and two observed product characteristics (x_{1jt} and x_{2jt}), p_{jt} is the price of the product, and ξ_{jt} and ϵ_{ijt} are unobservable shocks at the product-market and the individual product market level, respectively. The utility of the outside option is normalized to $u_{i0t} = \epsilon_{i0t}$. We draw each observed product characteristic x_{1jt} and x_{2jt} independently from the uniform distribution $U(0, 3)$, while ϵ_{ijt} is assumed to be distributed Type I extreme value. The distribution of the unobserved demand shocks ξ_{jt} is discussed below. The mean taste parameters β are set as $\beta = [1, 2, 1]$ while the price parameter $\alpha = -0.5$.

On the supply side, we assume that the marginal cost of producing product j in market t is $c_{jt} = \mathrm{w}_{jt}\gamma + \omega_{0jt}$ where w_{jt} is a vector containing a constant and two observed cost shifters (w_{1jt} and w_{2jt}) which are excluded from demand. Marginal cost also depends on ω_{0jt}, the true unobserved cost shock. As with the product characteristics, we draw each observed cost shifter w_{1jt} and w_{2jt} independently from the uniform distribution $U(0, 3)$. We adopt the default in PyBLP by drawing the unobserved demand and cost shocks ξ_{jt} and ω_{0jt} from a mean-zero bivariate normal distribution with variances of 1 and a correlation of 0.9. We set $\gamma = [3, 0.5, 1.5]$. For simplicity, the market size is normalized to one for all t. The government levies both a unit tax (τ_t) and an ad valorem tax (v_t) on all products in market t. The unit tax is remitted by the firms, while consumers remit the ad valorem tax. We draw the unit tax in each market from the uniform distribution $U(1, 2)$ while the ad valorem tax in each market is drawn from the uniform distribution $U(0, 0.2)$. We assume that the true model of conduct is Keystone pricing, whereby firms set tax-exclusive prices as twice their marginal cost, or $\nu_t p_{jt} = 2c_{jt} + 2\tau_t$.

D.1 Simulation Results

Test Results: Table 4 presents the results of testing all pairs of models. Examples 1-6 suggest that when testing Bertrand against Keystone, all instruments should be relevant.[39]

TABLE 4: Test Results for Simulated Data

Statistic	Instruments: Unit Tax (1)	Ad Valorem Tax (2)	Riv. Cost (3)	Riv. BLP (4)	Riv. BLP + Own x's (5)
Panel A: Bertrand vs. Keystone					
T^{RV}	25.2	16.1	9.8	16.3	29.4
	***	***	***	***	***
F	249.3	138.1	21.2	44.8	157.6
	††† $^{\wedge\wedge\wedge}$	††† $^{\wedge\wedge\wedge}$	††† $^{\wedge\wedge\wedge}$	††† $^{\wedge\wedge\wedge}$	††† $^{\wedge\wedge\wedge}$
Panel B: Cournot vs. Keystone					
T^{RV}	23.7	23.4	−0.0	0.7	40.8
	***	***			***
F	47.8	46.8	0.1	0.0	66.8
	††† $^{\wedge\wedge\wedge}$	††† $^{\wedge\wedge\wedge}$	†††	†††	††† $^{\wedge\wedge\wedge}$
Panel C: Shifted Keystone vs. Keystone					
T^{RV}	−0.1	4.8	0.6	0.5	0.5
		***			

The table reports, for each set of instruments and pair of models, the RV test statistics T^{RV}. Panels A and B also report the effective F-statistic (Duarte et al., 2024); we don't report it in Panel C as the pair of models does not satisfy Assumption 2 in Duarte et al. (2024). Columns 1-5 correspond to different instruments; panels A-C correspond to three different pairs of models. A positive RV test statistic suggests a better fit of Keystone. The symbol $***$ indicates rejection of the null of equal fit at the 0.01 significance level. The symbols $\dagger\dagger\dagger$ and $\wedge\wedge\wedge$ indicate that F is above the appropriate critical values for worst-case size below 0.075, and best-case power above 0.95, respectively. Both T^{RV} and the F-statistics account for market-level clustering. $n = 319,719$.

The results strongly support our theoretical predictions from Section 3. In Panel A, when testing Bertrand against Keystone, all instruments successfully reject the null in favor of Keystone. This aligns with our analysis showing that these models have different pass-through matrices. Under our logit demand specification, Bertrand exhibits positive off-diagonal pass-through elements (rival cost pass-through), while Keystone has a diagonal pass-through matrix. Our theoretical framework predicts that all instrument types should be relevant for this comparison: cost-side instruments target the ratios of off-diagonal to diagonal pass-through elements, demand-side instruments can target the full pass-through matrix, and tax instruments target various weighted combinations of pass-through elements along with level effects for ad valorem taxes.

[39] A table in Appendix 3.5 summarizes instrument relevance results for these examples.

Panel B reveals more nuanced patterns when testing Cournot against Keystone. Here, rival cost shifters (Column 3) and basic rival BLP instruments (Column 4) show virtually no power, with T^{RV} statistics close to zero and weak F-statistics. This confirms our theoretical prediction that both Cournot and Keystone have diagonal pass-through matrices under logit demand, making instruments that target off-diagonal pass-through elements irrelevant. As our analysis in Section 3 shows, rival cost shifters specifically target ratios of off-diagonal to diagonal pass-through elements. When both models have zero off-diagonal elements, this ratio is zero for both models, providing no identifying variation.

However, the inclusion of own product characteristics (Column 5) restores power, with the test decisively rejecting in favor of Keystone. This supports our finding that demand-side instruments can target the full pass-through matrix when they include own characteristics, not just the off-diagonal elements. The diagonal elements of the pass-through matrices differ between Cournot and Keystone ($\frac{s_{0t}}{1-s_{2t}}$ vs. 2 for firm 1), allowing falsification through own characteristic variation.

Tax instruments (Columns 1-2) also successfully distinguish Cournot from Keystone. Unit taxes target row sums of pass-through matrices, and since $P_{Ct}\iota \neq P_{Kt}\iota$ in our environment, this provides identifying power. Ad valorem taxes benefit from both the pass-through differences and the level effects, as the models have different markup levels.

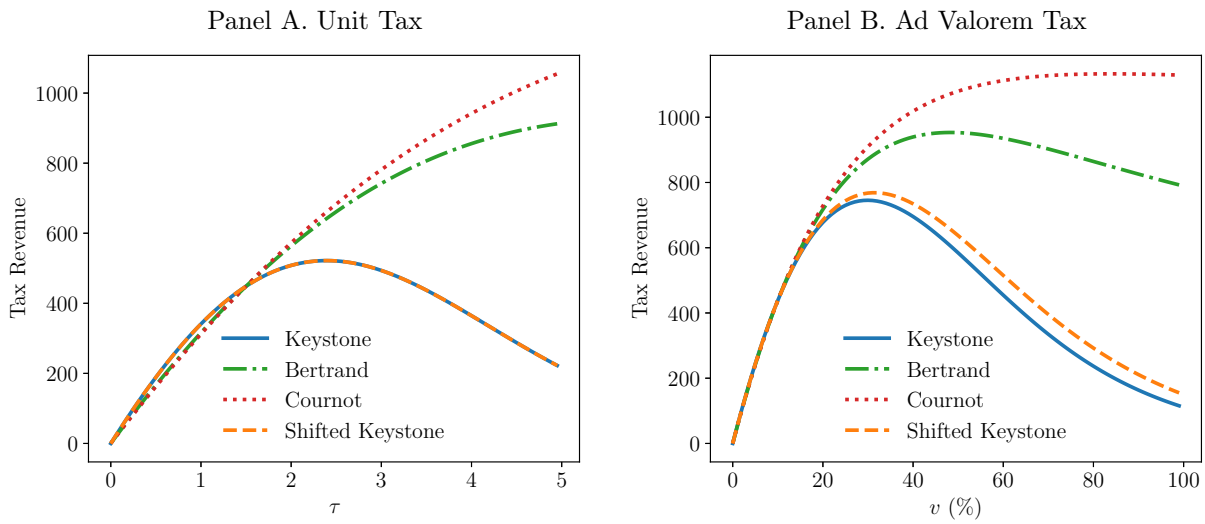
Panel C illustrates the power of ad valorem taxes to distinguish models through level effects rather than pass-through differences. The shifted Keystone model has identical pass-through to regular Keystone ($P_{St} = P_{Kt}$), making instruments that rely solely on pass-through differences irrelevant. Unit taxes, rival cost shifters, and product characteristics (Columns 1, 3-5) all show T^{RV} statistics near zero. However, ad valorem taxes (Column 2) strongly reject the shifted model, as they can detect differences in markup levels: $\Delta_{St} = \Delta_{Kt} + \eta$ where $\eta = 1$.

Laffer Curve: Next, we consider the tax implications of the models tested above. To mimic the problem facing a state government setting a single tax rate across many markets, we randomly select 500 of the 50,000 markets in our simulated data. In Panel A of Figure 7, we compute the Laffer curve for the unit tax, holding the ad valorem tax fixed at the levels in the simulated data. To obtain the Laffer curve under a given model of conduct, we first back out c_{mjt}, the implied marginal cost under that model. Holding demand and cost fixed, we then solve for equilibrium at each level of the tax and compute the associated revenue.

From the results in Table 4, we find that Keystone and shifted Keystone are the two models that are not falsified by unit tax instruments, as they feature the same pass-through matrices. Furthermore, in line with Remark 6, we see in Panel A of Figure 7 that these models imply the same Laffer curve. Conversely, since Bertrand and Cournot yield different

Laffer curves than the true one, they have different $P_{mt}\iota$ than Keystone, which permits the falsification of these models with unit tax instruments (Panels A and B of Table 4).

FIGURE 7: Laffer Curve for Unit and Ad Valorem Taxes



The figure illustrates the Laffer curve implied by different models of conduct. In Panel A, we hold fixed the ad valorem tax in each market and vary the unit tax from 0 to 5. In Panel B, we hold fixed the unit tax but vary the ad valorem tax from zero to one. To compute the curves in each panel, we simulate counterfactual prices, quantities, and government revenue at different levels of the appropriate tax for each candidate model.

In Panel B of Figure 7, we compute the Laffer curve for the ad valorem tax, holding the unit tax fixed at their simulated levels and following the same steps outlined for Panel A. In Table 1, we find that Keystone is the only model not falsified by ad valorem tax instruments. This is in line with Remark 6: the other models all produce a different Laffer curve, thus have different $P_{mt}(\Delta_{mt} - p_t)$, and are therefore falsified by ad valorem tax instruments.

Appendix E Additional Details: Data and Demand

E.1 Data Cleaning and Description

Table 5 summarizes the main data cleaning steps for our transaction-level data. We then take the cleaned transaction-level data and aggregate to the product-market level.

Table 5: Data Sample Cleaning Steps

		Resulting	
Step	Sample Restriction	Sample Size	Revenue ($)
0	Original	74,427,564	1,213,615,962
1	Keep usable products	53,883,838	813,353,991
2	Drop package size $\neq$ equal inventory usable weight	53,791,104	809,548,311
3	Keep 1 and 3.5g package sizes	34,418,816	490,387,068
4	Drop first 15 days of establishment's sales	34,286,246	488,555,343
5	keep $\frac{retailprices}{wholesaleprices} \in [1, 5]$	34,170,796	486,487,909
6	Keep retail weight ≤ 10	34,139,758	482,017,967

The table reports all steps to clean transaction-level data and their effect on sample size.

Table 6 provides descriptive statistics for the main variables in our database.

Table 6: Summary Statistics

	Mean	SD	Min	p25	Median	p75	Max
Price (per gram, tax inclusive)	11.11	3.14	4.54	9.02	10.93	12.92	21.07
Price (per gram, tax exclusive)	7.67	2.25	3.11	6.20	7.50	8.90	15.60
Shares (percent)	0.38	0.90	0.00	0.02	0.08	0.34	39.06
Wholesale Price (per gram)	3.63	0.92	1.50	3.00	3.60	4.12	6.50
Size	2.18	1.25	1.00	1.00	1.00	3.50	3.50
THC	5.42	7.94	0.00	0.53	0.99	7.06	62.66
CBD	0.42	1.54	0.00	0.01	0.10	0.23	36.93
Rival Products	364.25	544.76	0.00	24.00	116.00	429.00	2,109.00
Firms by Market	11.51	14.92	1.00	2.00	5.00	13.00	51.00
Processors by Market	94.89	77.64	1.00	34.00	69.00	134.00	316.00
Local Tax Rate	0.09	0.01	0.07	0.08	0.09	0.10	0.10
Observations	187,499						

The table reports summary statistics for the main variables in our database.

E.2 Demand Estimation

We provide here further details on the demand system introduced in Section 5.2. In our demand system, each consumer i receives utility from product j in market t according to the indirect utility:

$$u_{ijt} = x_{jt}\beta_i + \alpha_i p_{jt} + F_{r(j)} + F_{\ell(j)} + F_{m(t)} + \xi_{jt} + \zeta_{it} + (1-\rho)\epsilon_{ijt}$$

where x_j includes a constant, package size, THC and CBD (and their values squared), and the log of the number of products offered in the store; we include this variable to capture variation in shelf space across stores. The variable p_{jt} is the price of product j in market t, and $F_{r(j)}$, $F_{\ell(j)}$, $F_{m(t)}$ denote fixed effects for the retailer selling product j, the processor producing product j, and the year-month of the retail transaction, respectively. Consumer preferences for characteristics ($\beta_i = \bar{\beta} + \tilde{\beta} \times \text{Income}_i$) and price ($\alpha_i = \bar{\alpha} + \tilde{\alpha} \times \text{Income}_i$) vary with individual level income. ξ_{jt} and $\zeta_{it} + (1-\rho)\epsilon_{ijt}$ are unobservable shocks at the product-market and the individual product market level, respectively. Following the nested logit structure for our choice of nesting all inside goods together, ϵ_{ijt} is distributed Type 1 Extreme Value, and ζ_{it} is distributed according to the conjugate distribution (Cardell (1997)). To close the model we normalize consumer i's utility from the outside option as $u_{i0t} = \epsilon_{i0t}$. Given this utility specification, market shares s_{jt} as a function of observables, unobservables, and parameters take on the standard form (Berry et al. (1995); Grigolon and Verboven (2014)).

Identification and Estimation: The identifying assumption for our demand model is that, for a vector of demand instruments z^d_{jt}, the moment condition $E[\xi_{jt} z^d_{jt}] = 0$. We construct several demand instruments. We first construct the number of products sold at competing dispensaries in market t to help identify ρ. Following Gandhi and Houde (2025), we also interact this instrument with the mean income in the market to help identify income interaction parameters. We further construct BLP-style instruments to capture the closeness of products in the product space. Specifically, in a market, we sum the amount of THC and CBD both for products within a given store and also for products in all other stores. We also include exogenous own cost shifters, including the amount of rainfall and temperature in the region of production and their lags.

Because we do not take a stance on conduct, we perform demand estimation without using any supply-side moments.[40] However, we specify in Section 5.3 a menu of candidate models of conduct which includes Keystone pricing, Bertrand, and marginal cost pricing.

[40] See Appendix I of Duarte et al. (2024) for a comparison of sequential and simultaneous approaches to conduct testing.

The set of instruments specified above includes shifters of prices (e.g., own cost shifters) that will be relevant under any conduct model in our menu, including Keystone and marginal cost pricing.

Table 7: Demand Estimates

	(1) Logit-OLS		(2) Logit-2SLS		(3) RCNL	
	coef.	s.e.	coef.	s.e.	coef.	s.e.
Price (in $)	−0.1024	(0.0017)	−0.2831	(0.0418)	−0.4222	(0.0542)
Package Size (= 3.5 oz)	0.4414	(0.0066)	0.3352	(0.0254)	0.1012	(0.0441)
THC	0.0767	(0.0022)	0.0623	(0.0047)	0.0256	(0.0072)
THC Squared	−0.0034	(0.0001)	−0.0028	(0.0002)	−0.0011	(0.0003)
CBD	−0.0945	(0.0059)	−0.0837	(0.0093)	−0.044	(0.0097)
CBD Squared	−0.002	(0.0004)	−0.0022	(0.0007)	−0.0017	(0.0005)
Log Number Own Products	−4.835	(0.2647)	0.0012	(0.0173)	0.1281	(0.0282)
Constant						
ρ					0.3238	(0.0608)
Income × Constant					−0.0031	(0.0045)
Income × Price					0.0002	(0.0004)
Median Own Price Elasticity	−1.1134		−3.0792		−6.5219	
Median Aggregate Price Elasticity	−0.7966		−2.1922		−3.1617	
Diversion to outside option	0.6033		0.6421		0.4359	
Retailer FE	Yes		Yes		Yes	
Processor FE	Yes		Yes		Yes	
Year-Month FE	Yes		Yes		Yes	

The table reports demand estimates. Column 1: logit-OLS. Column 2: logit-2SLS. Column 3: full RCNL model. Income is measured in $100,000. $n = 187{,}499$.

Results: Results for demand estimation are reported in Table 7. For reference, we also report, in Columns 1 and 2, estimates from a simple logit model. In Column 1, we estimate the model with ordinary least squares. In Column 2, we estimate the model via 2SLS, using the same instruments as in the main specification. The reduction in the price coefficient between Columns 1 and 2 indicates the presence of endogeneity not controlled for by the fixed effects. Column 3 reports estimates of the full demand model. Compared to columns 1 and 2, the full model yields a more elastic demand system with diversion to the outside option that departs from logit.

Appendix F Robustness Checks

We present here additional robustness results for our empirical application.

F.1 Robustness to Box-Cox Transformations of Income

Following Miravete et al. (2026) we incorporate in our demand system a more flexible specification of income effects by adopting a Box-Cox transformation. Specifically, we allow consumer i's price sensitivity parameter α_i to depend on a nonlinear transformation of income (y_i), so that

$$\alpha_i = \bar{\alpha} + \tilde{\alpha} \times y_i^{\lambda-1}. \tag{6}$$

Our main specification corresponds to $\lambda = 2$. In principle, λ could be estimated as an additional parameter of the demand system. Because we lack the variation to credibly identify λ in our empirical environment, here we consider the robustness of our testing results to re-estimating our demand system for calibrated values of $\lambda \in \{0, 0.5, 1.5\}$. Table 8 reports demand estimates (in Panel A), and corresponding test results (in Panel B) for the three values of λ. While the nonlinear transformation of income makes a difference for some features of the demand system, test results are consistent with our main specification in Section 5, and strongly support Keystone as the better fitting model.

Table 8: Robustness to Box-Cox Transformations of Income in Demand

Panel A: Demand Estimates						
	RCNL Box-Cox					
	$\lambda = 0$		$\lambda = 0.5$		$\lambda = 1.5$	
	coef.	s.e.	coef.	s.e.	coef.	s.e.
Price (in \$)	−0.400	(0.041)	−0.399	(0.042)	−0.450	(0.070)
Package Size (= 3.5 oz)	0.098	(0.043)	0.094	(0.044)	0.097	(0.044)
THC	0.025	(0.007)	0.024	(0.007)	0.025	(0.007)
THC Squared	−0.001	(0.000)	−0.001	(0.000)	−0.001	(0.000)
CBD	−0.043	(0.009)	−0.043	(0.009)	−0.044	(0.010)
CBD Squared	−0.002	(0.001)	−0.002	(0.001)	−0.002	(0.001)
Log Number Own Products	0.135	(0.026)	0.132	(0.026)	0.127	(0.027)
ρ	0.342	(0.053)	0.336	(0.054)	0.322	(0.058)
Income × Constant	−0.000	(0.001)	−0.001	(0.002)	−0.003	(0.004)
Income × Price	0.076	(0.416)	−0.073	(0.184)	0.004	(0.005)
Median Own Price Elasticity	−6.562		−6.696		−6.626	
Median Aggregate Price Elasticity	−3.071		−3.165		−3.229	
Diversion to outside option	0.424		0.428		0.437	
Panel B: Testing Results – Bertrand vs. Keystone, Tax Instruments						
T^{RV}	13.317		13.309		13.273	
	$***$		$***$		$***$	
F	63.067		61.849		57.562	
	$\dagger\dagger\dagger$ $\wedge\wedge\wedge$		$\dagger\dagger\dagger$ $\wedge\wedge\wedge$		$\dagger\dagger\dagger$ $\wedge\wedge\wedge$	

The table reports demand estimates (Panel A) and conduct test results (Panel B) for Box-Cox transformations of income (measured in \$100,000). Columns correspond to different calibrated values of the λ parameter in Equation (6). All demand specifications include retailer, processor, and year-month fixed effects. Panel B reports, for each value of λ, the RV test statistic T^{RV} and the effective F-statistic (Duarte et al., 2024) for testing Bertrand versus Keystone with ad valorem tax instruments. The symbol $***$ indicates rejection of the null of equal fit at the 0.01 significance level. The symbols $\dagger\dagger\dagger$ and $\wedge\wedge\wedge$ indicate that F is above the appropriate critical values for worst-case size below 0.075, and best-case power above 0.95, respectively. Both T^{RV} and the F-statistics account for two-step estimation error and clustering at the market level. $n = 187{,}499$.

F.2 Robustness of Test Results to Alternative Cost Specifications

We consider the robustness of our testing results in Table 3 to alternative specifications of marginal cost. Results are reported in Table 9. Panel A reproduces the specification in the main text. Panels B and C include package size indicators and consider alternative specifi-

cations of fixed effects in marginal cost. For all specifications, our preferred tax instruments are strong for size and power, and conclude for superior fit of the Keystone model. Different specifications of marginal cost affect the variation that is available for testing conduct. In particular, the specifications in Panels B and C incorporate both geographic and time fixed effects, thus absorbing considerable variation. This weakens instruments, especially RC and PC. Despite these additional hurdles, whenever the null is rejected, the test always rejects in favor of Keystone.

Table 9: Test Results for Different Levels of Fixed Effects

Statistic	Instruments: Tax (1)	RC (2)	PC (3)	WP (4)
Panel A: Product Fixed Effects				
T^{RV}	13.519	3.767	9.555	17.037
	***	***	***	***
F	34.338	2.126	18.653	1,639.6
	††† $^{\wedge\wedge\wedge}$	††† $^{\wedge\wedge\wedge}$	††† $^{\wedge\wedge\wedge}$	††† $^{\wedge\wedge\wedge}$
Panel B: Retailer, Processor, and Year-Month Fixed Effects				
T^{RV}	2.182	−0.153	2.249	9.603
	***		***	***
F	4.663	0.599	1.534	3,665.8
	††† $^{\wedge\wedge\wedge}$	†††	††† $^{\wedge\wedge\wedge}$	††† $^{\wedge\wedge\wedge}$
Panel C: Retailer, and Year-Month Fixed Effects				
T^{RV}	2.521	−0.109	2.422	25.718
	***		***	***
F	4.521	0.597	1.515	2,398.3
	††† $^{\wedge\wedge\wedge}$	†††	††† $^{\wedge\wedge\wedge}$	††† $^{\wedge\wedge\wedge}$

The table reports, for each set of instruments, the RV test statistics T^{RV} and the effective F-statistic (Duarte et al., 2024) for testing Bertrand versus Keystone. Different panels correspond to different levels of fixed effects in marginal cost. A positive RV test statistic suggests a better fit of Keystone. The symbol $***$ indicates rejection of the null of equal fit at the 0.01 significance level. The symbols $\dagger\dagger\dagger$ and $\wedge\wedge\wedge$ indicate that F is above the appropriate critical values for worst-case size below 0.075, and best-case power above 0.95, respectively. Both T^{RV} and the F-statistics account for two-step estimation error and clustering at the market level. $n = 187,499$.

F.3 Robustness of Test Results to Alternative Price Parameters

We consider the robustness of our testing results in Table 3 to alternative values of the median aggregate own price elasticity (-3.16 for our main specification). Here we lower the magnitude of elasticities in steps of one to -2.16 and -1.16. For each, we take the prices and shares in the data and calibrate demand by adjusting the price parameter α, holding the other demand parameter fixed at the estimates reported in Column 3 of Table 7. Results are reported in Table 10.[41] The test results are robust at each of the aggregate price elasticities we consider – all four instrument sets are strong for both size and power and reject the null of equal fit in favor of the Keystone model.

Table 10: Test Results for Different Price Parameters

Statistic	Instruments: Tax (1)	RC (2)	PC (3)	WP (4)
Panel A: $\alpha = -0.291$, median aggregate own-price elasticity $= -2.16$				
T^{RV}	13.159	5.586	10.727	17.133
	$***$	$***$	$***$	$***$
F	49.182	1.106	17.164	2,166.3
	$\dagger\dagger\dagger$ $\wedge\wedge\wedge$	$\dagger\dagger\dagger$ $\wedge\wedge\wedge$	$\dagger\dagger\dagger$ $\wedge\wedge\wedge$	$\dagger\dagger\dagger$ $\wedge\wedge\wedge$
Panel B: $\alpha = -0.162$, median aggregate own-price elasticity $= -1.16$				
T^{RV}	12.096	4.539	11.191	16.738
	$***$	$***$	$***$	$***$
F	28.889	2.735	21.935	1,543.4
	$\dagger\dagger\dagger$ $\wedge\wedge\wedge$	$\dagger\dagger\dagger$ $\wedge\wedge\wedge$	$\dagger\dagger\dagger$ $\wedge\wedge\wedge$	$\dagger\dagger\dagger$ $\wedge\wedge\wedge$

The table reports, for each set of instruments, the RV test statistic T^{RV} and the effective F-statistic (Duarte et al., 2024) for testing Bertrand versus Keystone. Different panels correspond to different price parameters chosen to imply median aggregate price elasticities of -2.16 and -1.16, from top to bottom. A positive RV test statistic suggests a better fit of Keystone. The symbol $***$ indicates rejection of the null of equal fit at the 0.01 significance level. The symbols $\dagger\dagger\dagger$ and $\wedge\wedge\wedge$ indicate that F is above the appropriate critical values for worst-case size below 0.075, and best-case power above 0.95, respectively. Both T^{RV} and the F-statistics account for clustering at the market level. $n = 187{,}499$.

[41] Given that we are simulating demand, the RV test statistic and the $F-$ statistic are only adjusted for clustering, not to account for uncertainty stemming from demand estimation. Test statistics for the main specification reported in Table 3 are only slightly changed.

Appendix References

Backus, M., C. Conlon, and M. Sinkinson (2021): "Common Ownership and Competition in the Ready-To-Eat Cereal Industry," NBER Working Paper #28350.

Berry, S. and P. Haile (2014): "Identification in Differentiated Products Markets Using Market Level Data," *Econometrica*, 82, 1749–1797.

Berry, S., J. Levinsohn, and A. Pakes (1995): "Automobile Prices in Market Equilibrium," *Econometrica*, 63, 841–890.

Cardell, N. (1997): "Variance Components Structures for the Extreme-Value and Logistic Distributions with Application to Models of Heterogeneity," *Econometric Theory*, 13, 185–213.

Duarte, M., L. Magnolfi, M. Sølvsten, and C. Sullivan (2024): "Testing Firm Conduct," *Quantitative Economics*, 15, 571–606.

Gandhi, A. and J.-F. Houde (2025): "Measuring Substitution Patterns in Differentiated Products Industries," Working Paper.

Grigolon, L. and F. Verboven (2014): "Nested Logit or Random Coefficients Logit? A Comparison of Alternative Discrete Choice Models of Product Differentiation," *Review of Economics and Statistics*, 96, 916–935.

Magnolfi, L. and C. Sullivan (2022): "A Comparison of Testing and Estimation of Firm Conduct," *Economics Letters*, 212, 110316.

Miller, N. and M. Weinberg (2017): "Understanding the Price Effects of the MillerCoors Joint Venture," *Econometrica*, 85, 1763–1791.